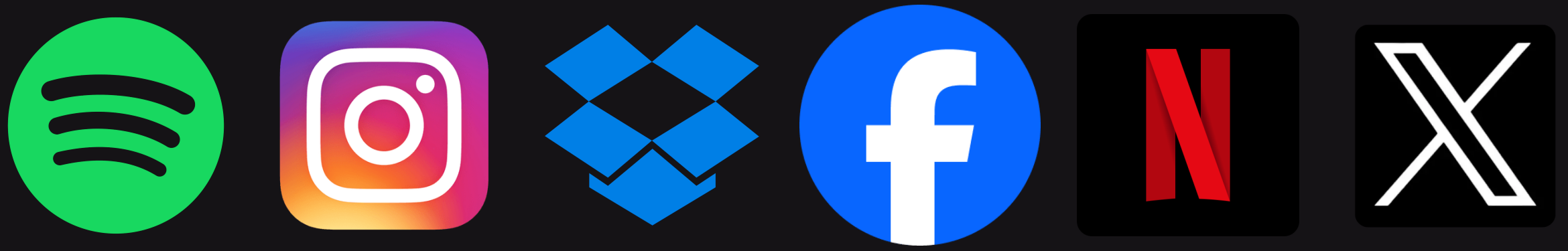
Analytics Vidhya

The Exponential Growth

Instructor

Kunal Jain

Founder and CEO, Analytics Vidhya



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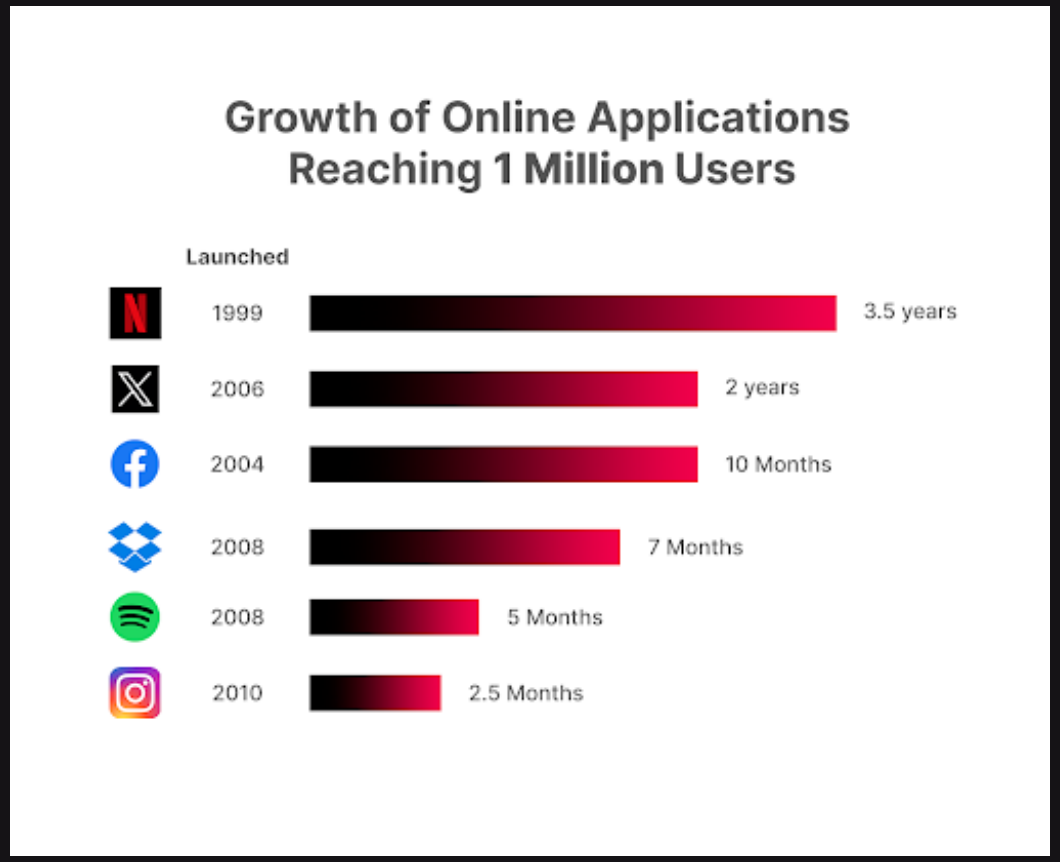
Growth of Online Applications!



Time it took for these online users to reach 1 million users

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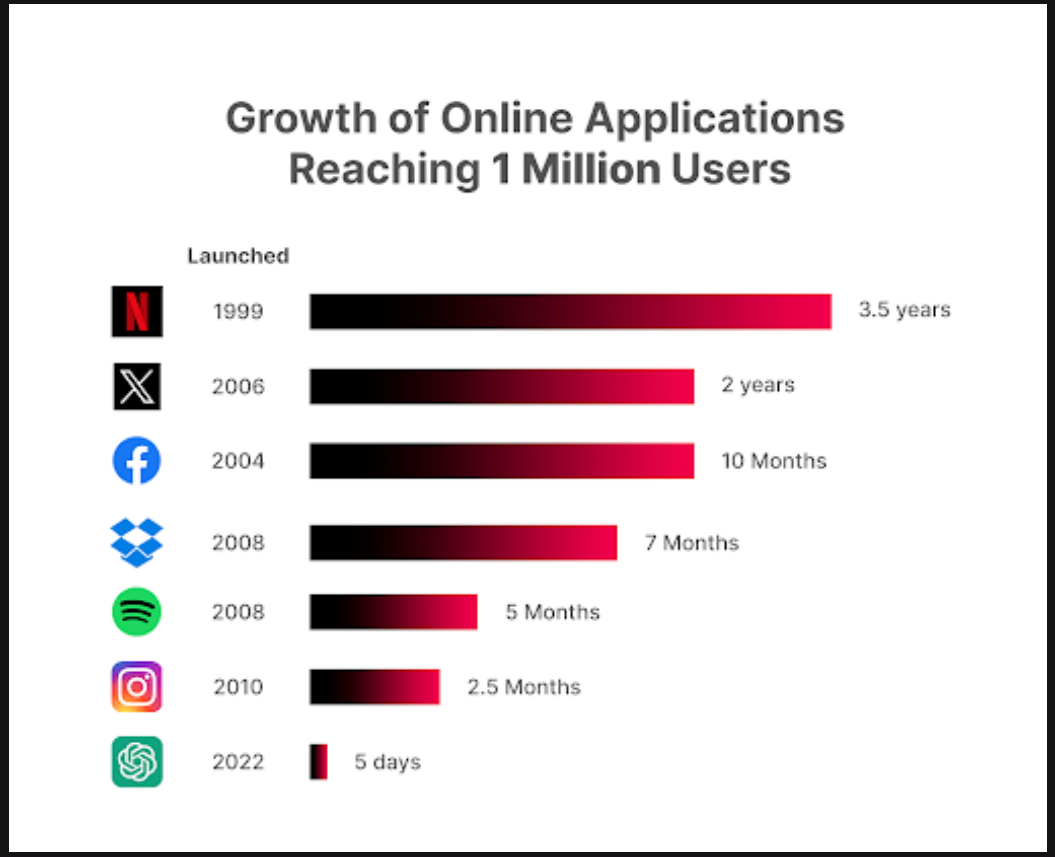
Growth of ChatGPT!



Time it took for these online users to reach 1 million users

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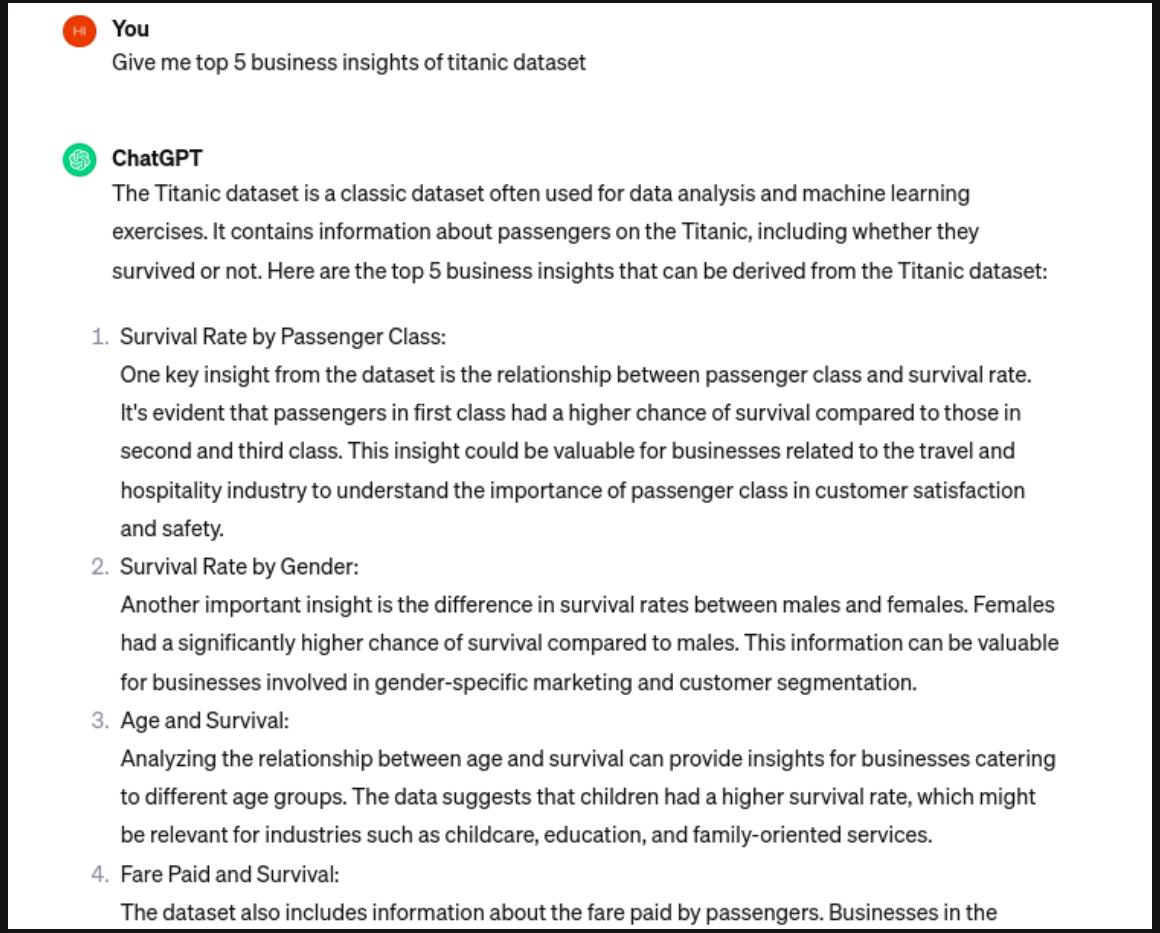
ChatGPT for Data Analysts

- Analyze datasets and generate the business insights
- Derive meaningful information and trends in seconds



You

Give me top 5 business insights of titanic dataset

ChatGPT

The Titanic dataset is a classic dataset often used for data analysis and machine learning exercises. It contains information about passengers on the Titanic, including whether they survived or not. Here are the top 5 business insights that can be derived from the Titanic dataset:

1. Survival Rate by Passenger Class:
   One key insight from the dataset is the relationship between passenger class and survival rate. It's evident that passengers in first class had a higher chance of survival compared to those in second and third class. This insight could be valuable for businesses related to the travel and hospitality industry to understand the importance of passenger class in customer satisfaction and safety.
2. Survival Rate by Gender:
   Another important insight is the difference in survival rates between males and females. Females had a significantly higher chance of survival compared to males. This information can be valuable for businesses involved in gender-specific marketing and customer segmentation.
3. Age and Survival:
   Analyzing the relationship between age and survival can provide insights for businesses catering to different age groups. The data suggests that children had a higher survival rate, which might be relevant for industries such as childcare, education, and family-oriented services.
4. Fare Paid and Survival:
   The dataset also includes information about the fare paid by passengers. Businesses in the

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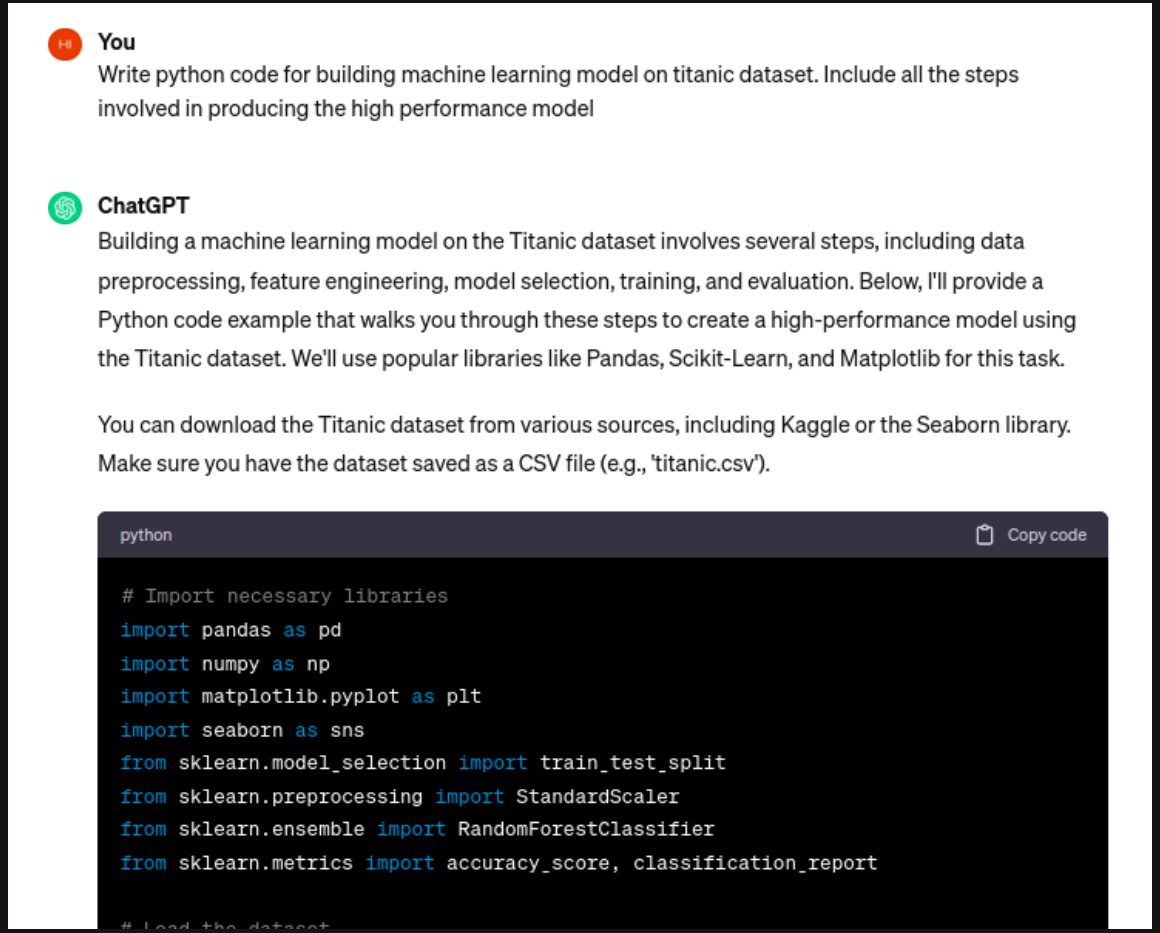
ChatGPT for Data Scientists

- Automate machine learning code development
- Eliminates the need to remember any form of code



You

Write python code for building machine learning model on titanic dataset. Include all the steps involved in producing the high performance model

ChatGPT

Building a machine learning model on the Titanic dataset involves several steps, including data preprocessing, feature engineering, model selection, training, and evaluation. Below, I'll provide a Python code example that walks you through these steps to create a high-performance model using the Titanic dataset. We'll use popular libraries like Pandas, Scikit-Learn, and Matplotlib for this task.

You can download the Titanic dataset from various sources, including Kaggle or the Seaborn library. Make sure you have the dataset saved as a CSV file (e.g., 'titanic.csv').

python Copy code

```python
# Import necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report

# Load the dataset
```

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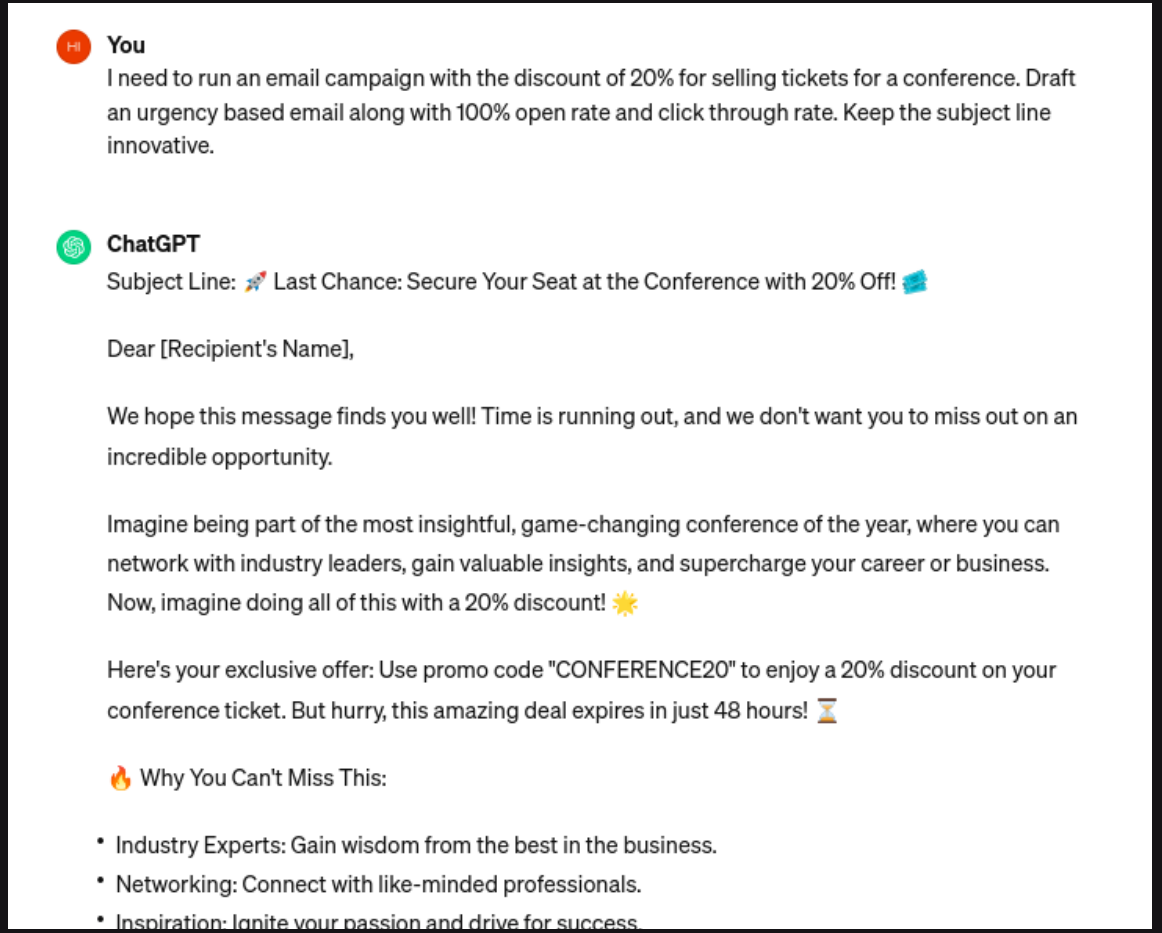
ChatGPT for Sales and Marketing

- Craft compelling emails for their target audience
- Increase the conversions and leads through email campaigns



You

I need to run an email campaign with the discount of 20% for selling tickets for a conference. Draft an urgency based email along with 100% open rate and click through rate. Keep the subject line innovative.

ChatGPT

Subject Line: 🚀 Last Chance: Secure Your Seat at the Conference with 20% Off! 🎟️

Dear [Recipient's Name],

We hope this message finds you well! Time is running out, and we don't want you to miss out on an incredible opportunity.

Imagine being part of the most insightful, game-changing conference of the year, where you can network with industry leaders, gain valuable insights, and supercharge your career or business. Now, imagine doing all of this with a 20% discount! 🌟

Here's your exclusive offer: Use promo code "CONFERENCE20" to enjoy a 20% discount on your conference ticket. But hurry, this amazing deal expires in just 48 hours! ⏳

🔥 Why You Can't Miss This:

- Industry Experts: Gain wisdom from the best in the business.
- Networking: Connect with like-minded professionals.
- Inspiration: Ignite your passion and drive for success.

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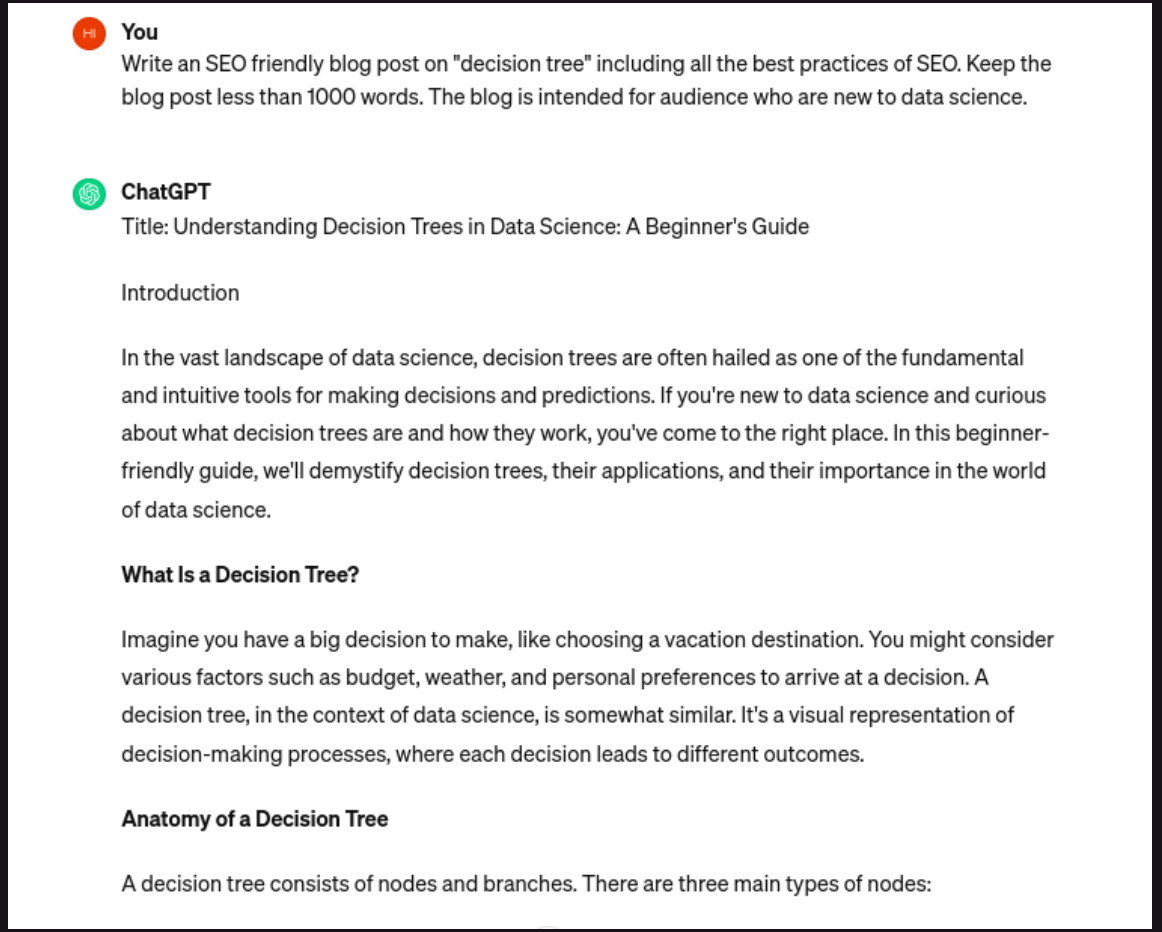
ChatGPT for Content Creators

- Generate engaging content quickly
- Generate blog posts, social media posts, video scripts etc.



You

Write an SEO friendly blog post on "decision tree" including all the best practices of SEO. Keep the blog post less than 1000 words. The blog is intended for audience who are new to data science.

ChatGPT

Title: Understanding Decision Trees in Data Science: A Beginner's Guide

Introduction

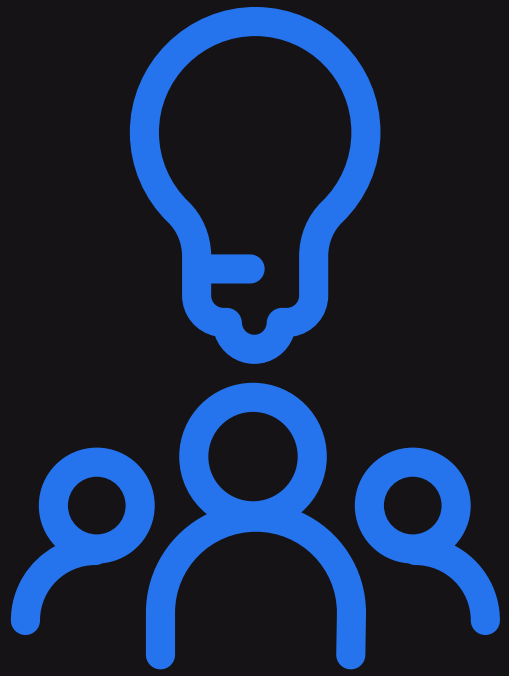
In the vast landscape of data science, decision trees are often hailed as one of the fundamental and intuitive tools for making decisions and predictions. If you're new to data science and curious about what decision trees are and how they work, you've come to the right place. In this beginner-friendly guide, we'll demystify decision trees, their applications, and their importance in the world of data science.

What Is a Decision Tree?

Imagine you have a big decision to make, like choosing a vacation destination. You might consider various factors such as budget, weather, and personal preferences to arrive at a decision. A decision tree, in the context of data science, is somewhat similar. It's a visual representation of decision-making processes, where each decision leads to different outcomes.

Anatomy of a Decision Tree

A decision tree consists of nodes and branches. There are three main types of nodes:



Our Imagination is the only limit!

Research Explosion

Research & Experimentation

Large Language Models - LLMs

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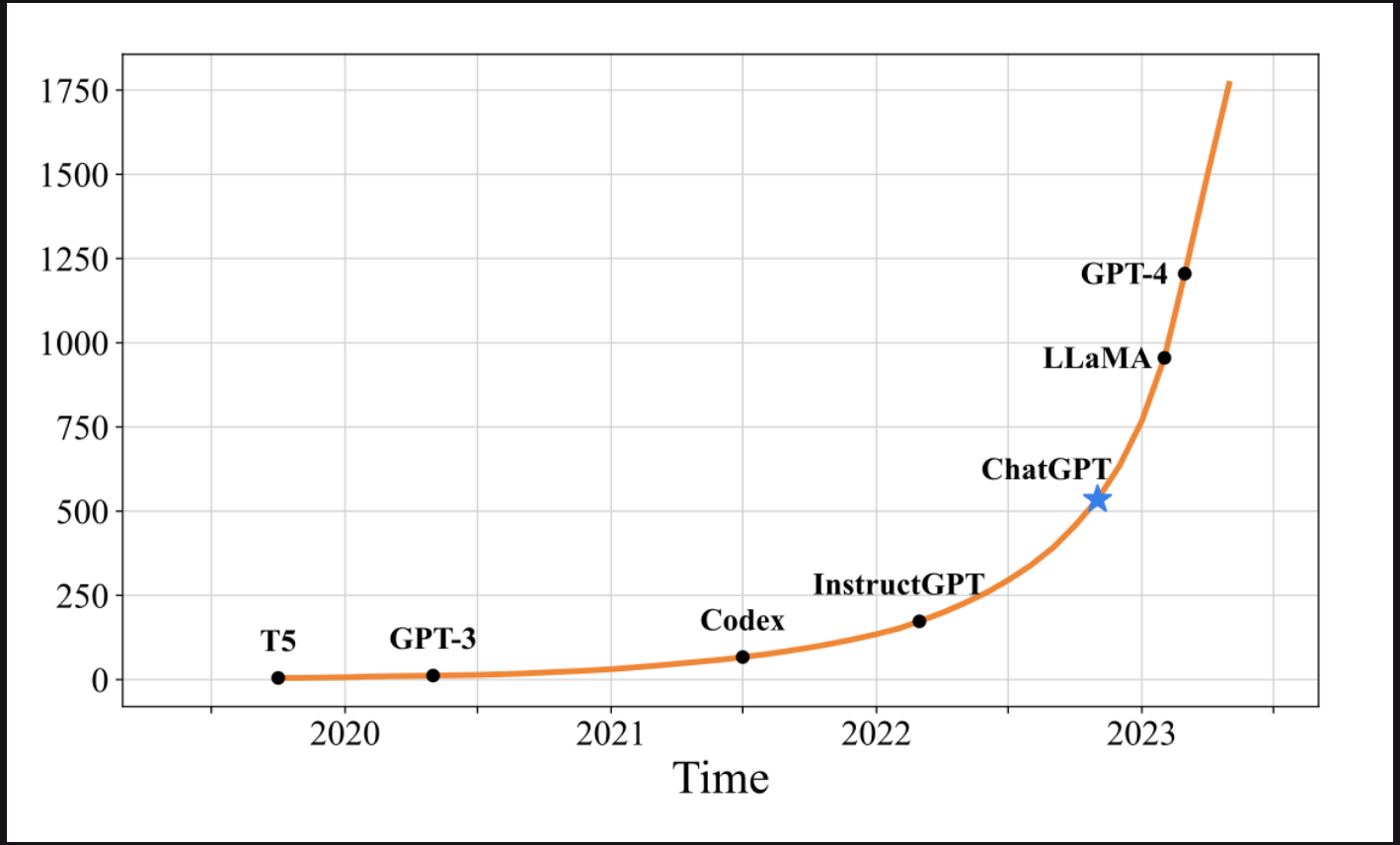
Research Explosion



No. of cumulative arxiv papers containing keyphrase “Large Language Model” over the years

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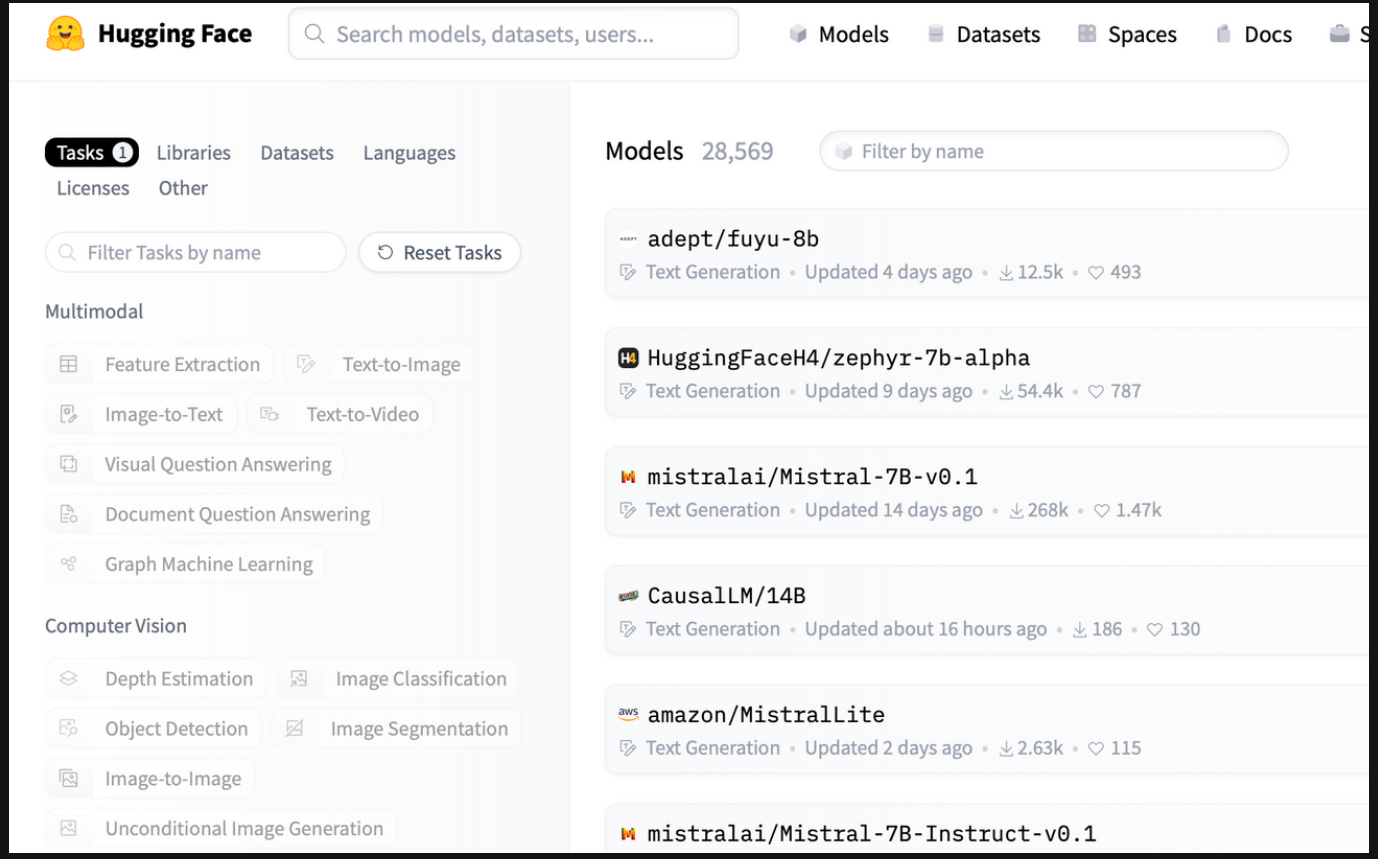
Community Contribution



No. of “text generation” models on Hugging Face rise to 28500+ till Nov 2023

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The Evolution of NLP

Develop a system that helps computers understand, interpret, and communicate with human in human languages

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The Evolution of NLP

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The Evolution of NLP: Symbolic NLP

Instructor

Kunal Jain

Founder and CEO, Analytics Vidhya

NLP Evolution

1950 - 1980

Symbolic NLP

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NLP system used complex hand written rules to understand and communicate with human aka **Symbolic NLP**

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Georgetown-IBM Experiment (1954)

The first public demonstration of machine translation: the Georgetown-IBM system, 7th January 1954

John Hutchins[1]

Abstract

The public demonstration of a Russian-English machine translation system in New York in January 1954 – a collaboration of IBM and Georgetown University – caused a great deal of public interest and much controversy. Although a small-scale experiment of just 250 words and six 'grammar' rules it raised expectations of automatic systems capable of high quality translation in the near future. This paper describes the background motivations, the linguistic methods, and the computational techniques of the system.

1. Introduction

On the 8th January 1954, the front pages of the *New York Times* and other major American newspapers (*New York Herald Tribune, Christian Science Monitor, Washington Herald Tribune, Los Angeles Times*) carried reports of the first public demonstration of a computer for translating languages. Reports were syndicated in many provincial newspapers, and articles about it appeared in the following months in popular magazines (*Newsweek, Science, Science News Letter, Discovery, Chemical Week, Chemical Engineering News, Electrical Engineering, Mechanical World, Computers and Automation*, etc.) It was probably the most widespread and influential publicity that machine translation (MT) has ever received,[2] and it was undoubtedly the first non-numerical application of the newly invented 'electronic brains' that most people had heard of. Translation itself was a largely unknown 'art' and the prospect of a machine capable of 'deciphering' foreign languages was exciting. The demonstration raised expectations of fast and easy international communication in a world that had already become divided by confrontations and misunderstandings between the United States and the Soviet Union.

"ELIZA"

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Eliza (1967)

```
  Welcome to
                 EEEEEE  LL      IIII   ZZZZZZ   AAAAA
                 EE      LL       II       ZZ   AA   AA
                 EEEEE   LL       II     ZZZ    AAAAAAA
                 EE      LL       II    ZZ      AA   AA
                 EEEEEE  LLLLLL  IIII ZZZZZZ    AA   AA

  Eliza is a mock Rogerian psychotherapist.
  The original program was described by Joseph Weizenbaum in 1966.
  This implementation by Norbert Landsteiner 2005.


ELIZA: Is something troubling you ?
YOU:   Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU:   They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
YOU:   Well, my boyfriend made me come here.
ELIZA: Is it important to you that your boyfriend made you come here ?
YOU:   He says I'm depressed much of the time.
ELIZA: I am sorry to hear that you are depressed.
YOU:   It's true. I am unhappy.
ELIZA: Can you explain what made you unhappy ?
YOU:
```

SHRDLU (1968)

- An improved version of the chatbot compared to Eliza
- Approach: Predefined Vocabulary
- Limitation: Restricted to set of words and context

Challenge: Symbolic Models were not accurate and scalable

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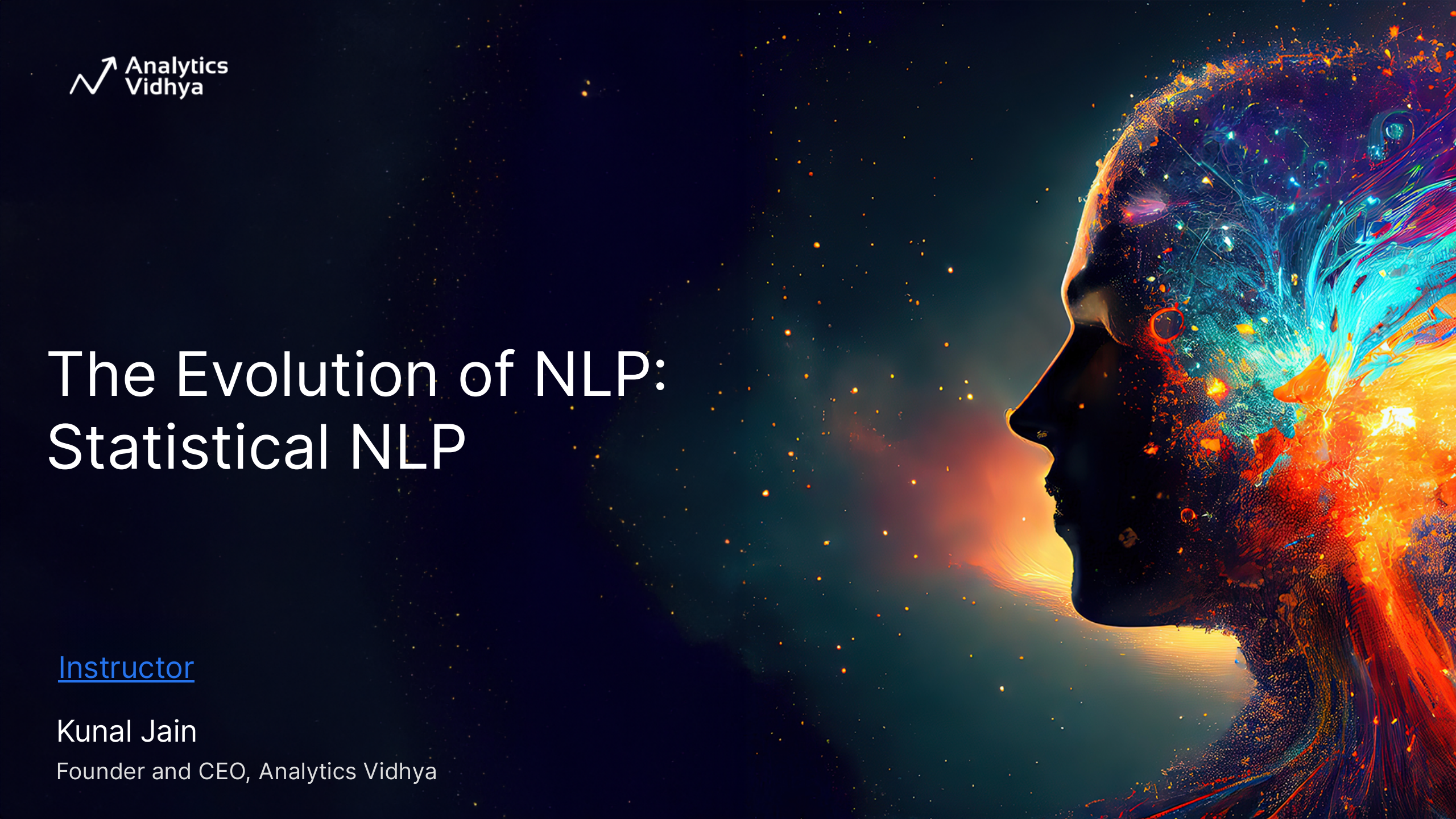
The Evolution of NLP: Statistical NLP



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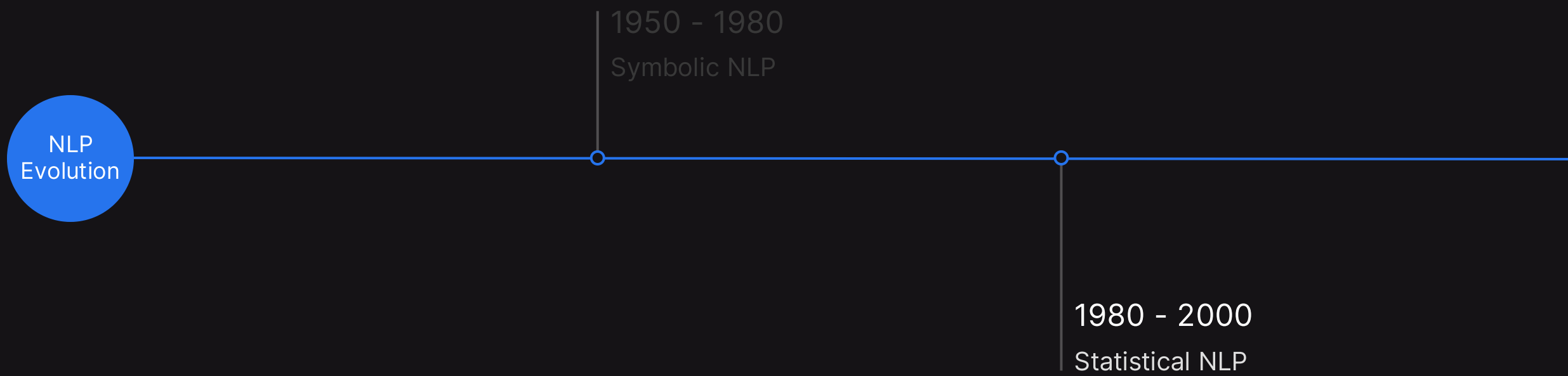

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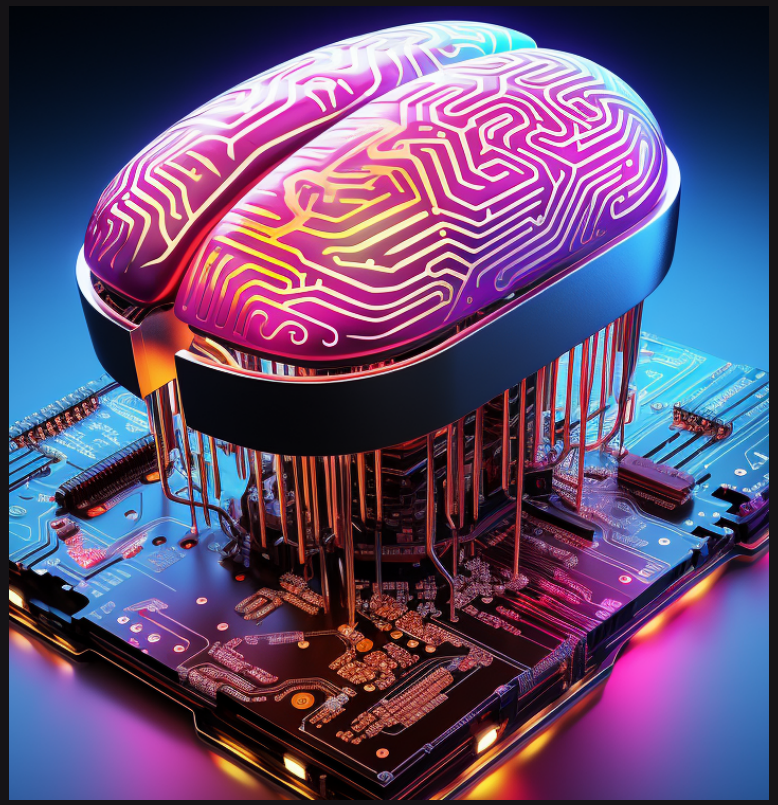
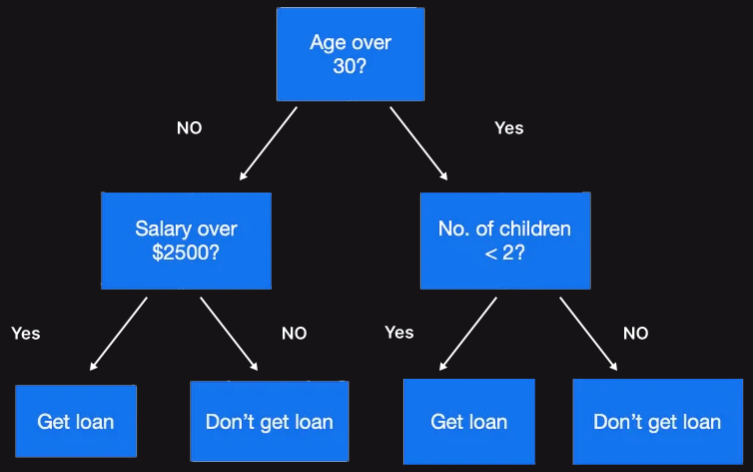
Growth in NLP

Compute Power



Applied Machine Learning Algorithms



Decision Tree

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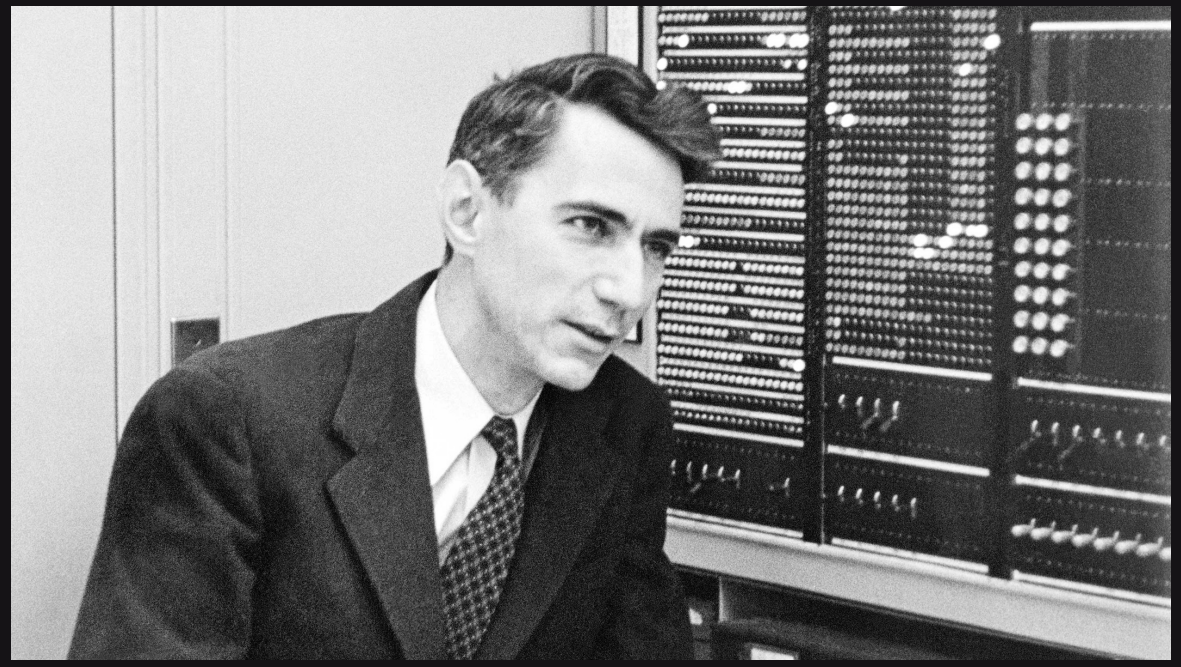
N-Gram Language Model



C.E Shannon

Prediction and Entropy of Printed English

By C. E. SHANNON

(*Manuscript Received Sept. 15, 1950*)

A new method of estimating the entropy and redundancy of a language is described. This method exploits the knowledge of the language statistics possessed by those who speak the language, and depends on experimental results in prediction of the next letter when the preceding text is known. Results of experiments in prediction are given, and some properties of an ideal predictor are developed.

1. INTRODUCTION

IN A previous paper[1] the entropy and redundancy of a language have been defined. The entropy is a statistical parameter which measures, in a certain sense, how much information is produced on the average for each letter of a text in the language. If the language is translated into binary digits (0 or 1) in the most efficient way, the entropy H is the average number of binary digits required per letter of the original language. The redundancy, on the other hand, measures the amount of constraint imposed on a text in the language due to its statistical structure, e.g., in English the high frequency of the letter E, the strong tendency of H to follow T or of V to follow Q. It was estimated that when statistical effects extending over not more

"Prediction and Entropy of Printed English"

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N-Gram Language Model

- Predicts next word in the sentence
- For example, the boy is going to .....
- The next set of possible words can be school, home, play
- Probabilities: play (0.01), home (0.6), school (0.8)
- Next word: school

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Computing Probabilities

Probability is computed based on the no. of occurrences of a word in the training data.

Training Data

The boy is going to school

The boy is going to play

The boy is going to dance

The boy is going to home

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Computing Probabilities

Word	Probabilities
p(school/The boy is going to)	2/5
p(play/The boy is going to)	1/5
p(dance/The boy is going to)	1/5
p(home/The boy is going to)	1/5

Next word will be “school”

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Limitations of N-Gram Language Model

- It cannot predict for new sentences.
  "The boy is watching...?..."
- It is deterministic in nature.

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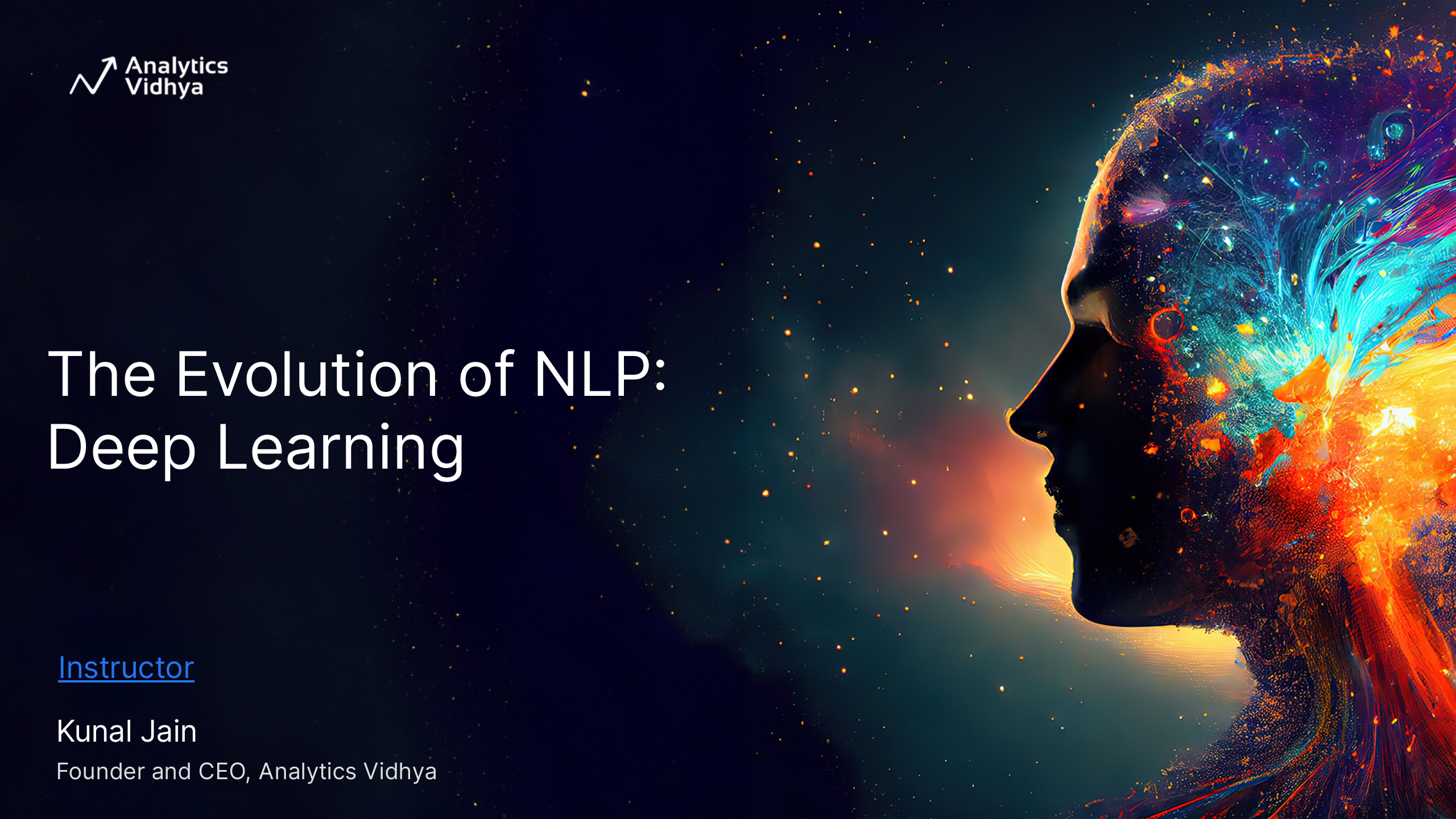
The Evolution of NLP: Deep Learning



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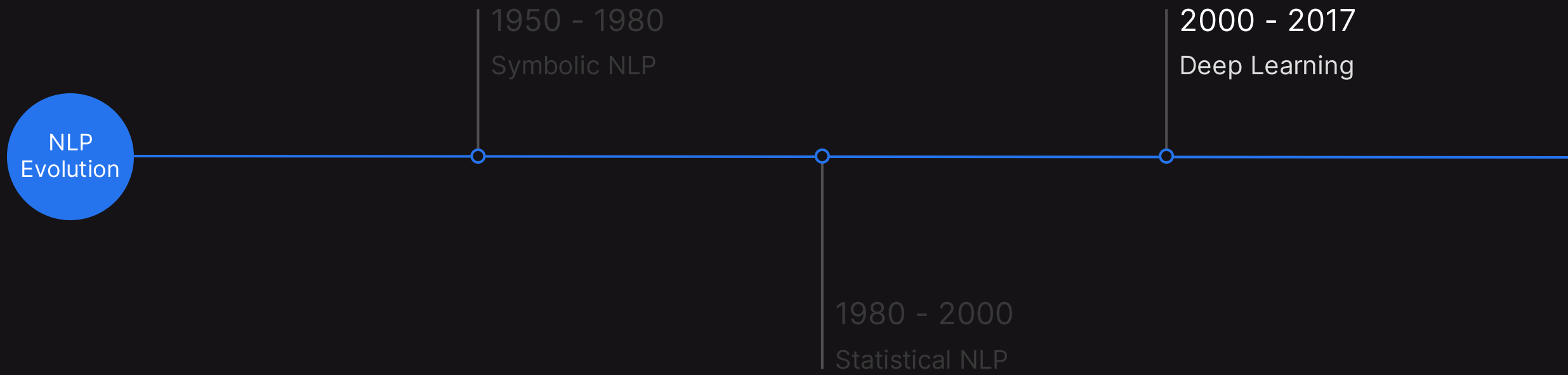
Founder and CEO, Analytics Vidhya



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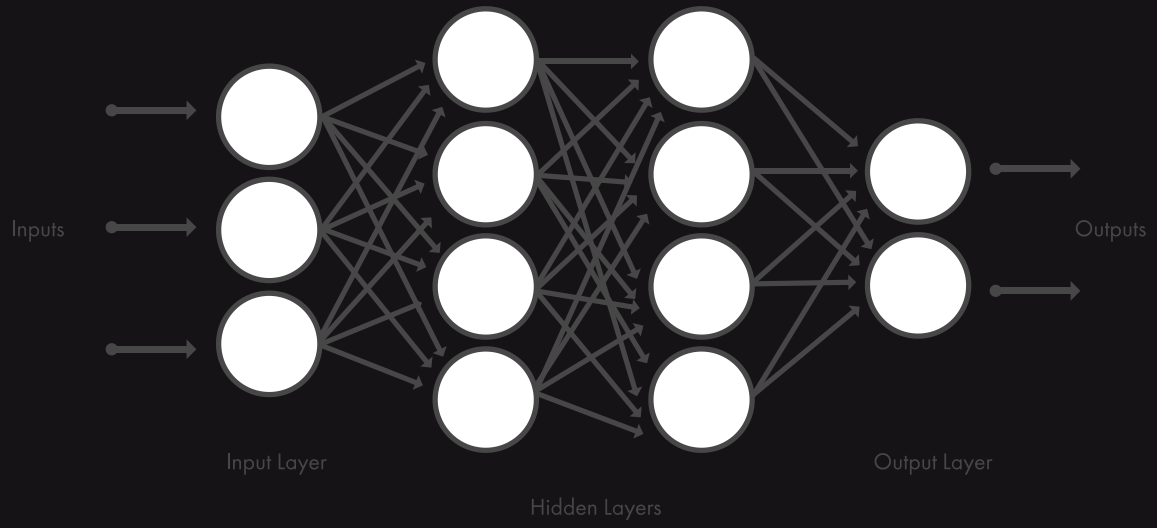
Growth in NLP

Massive Compute Power



Deep Learning

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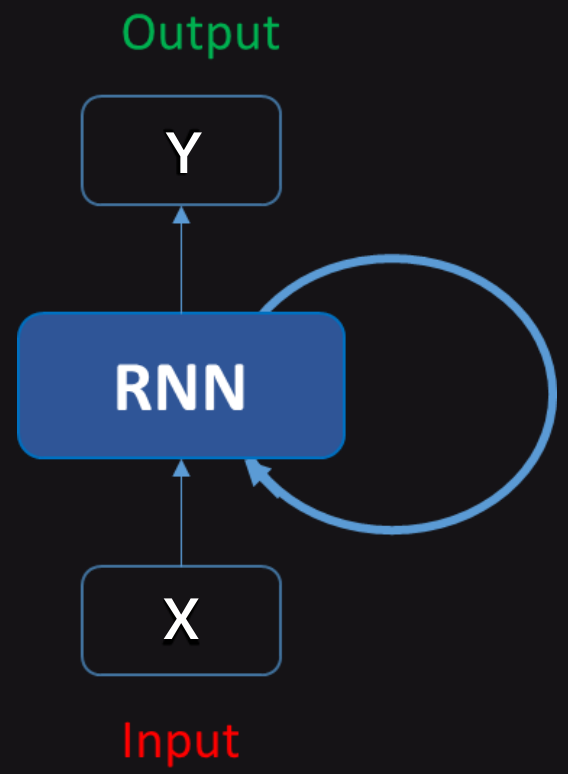
Recurrent Neural Network



Recurrent Neural Network

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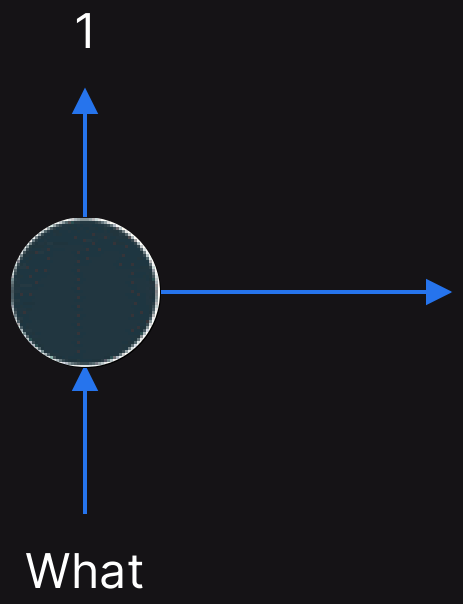
Recurrent Neural Network



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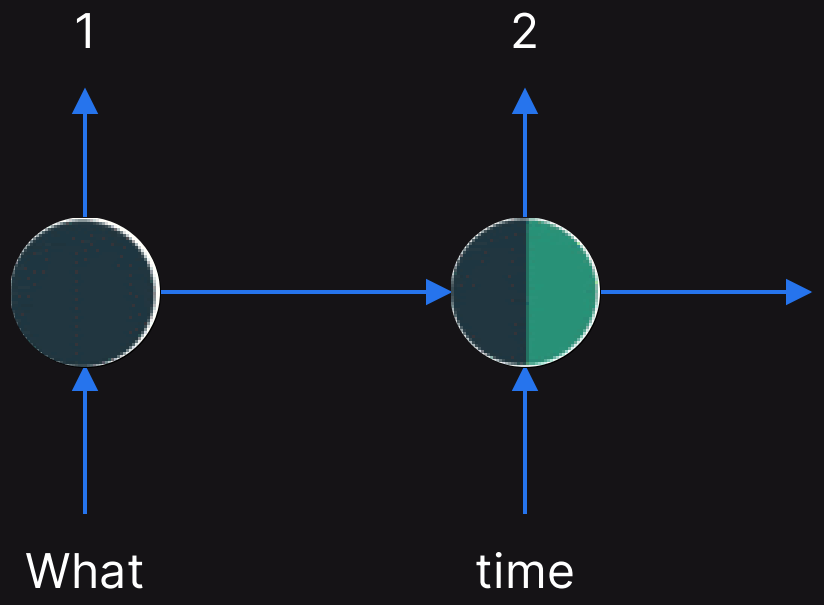
Recurrent Neural Network



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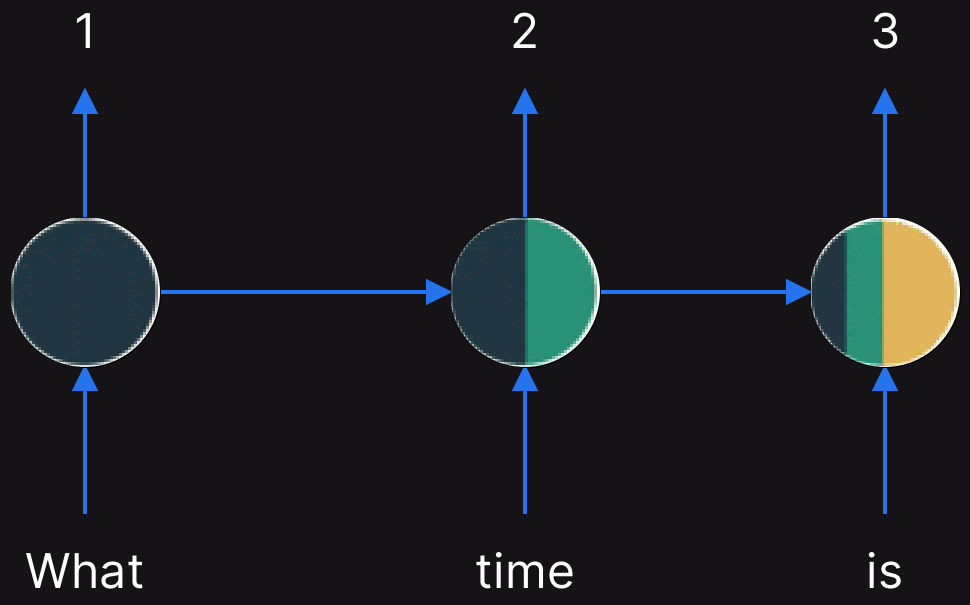
Recurrent Neural Network



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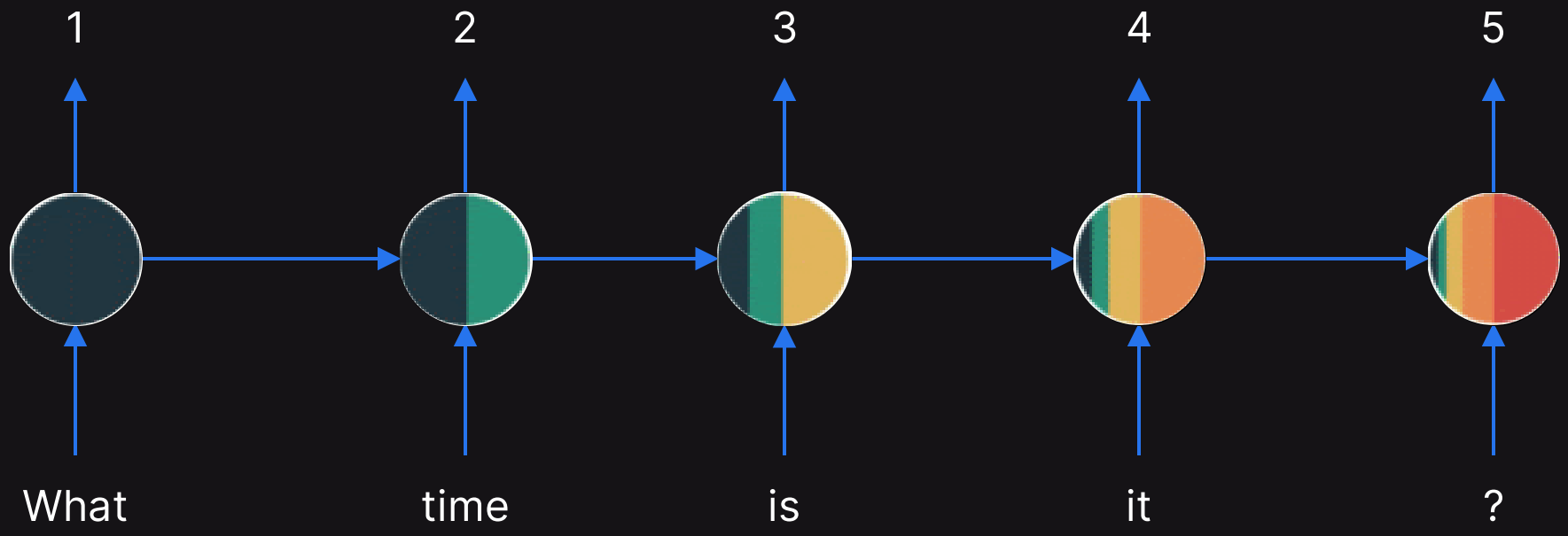
Recurrent Neural Network



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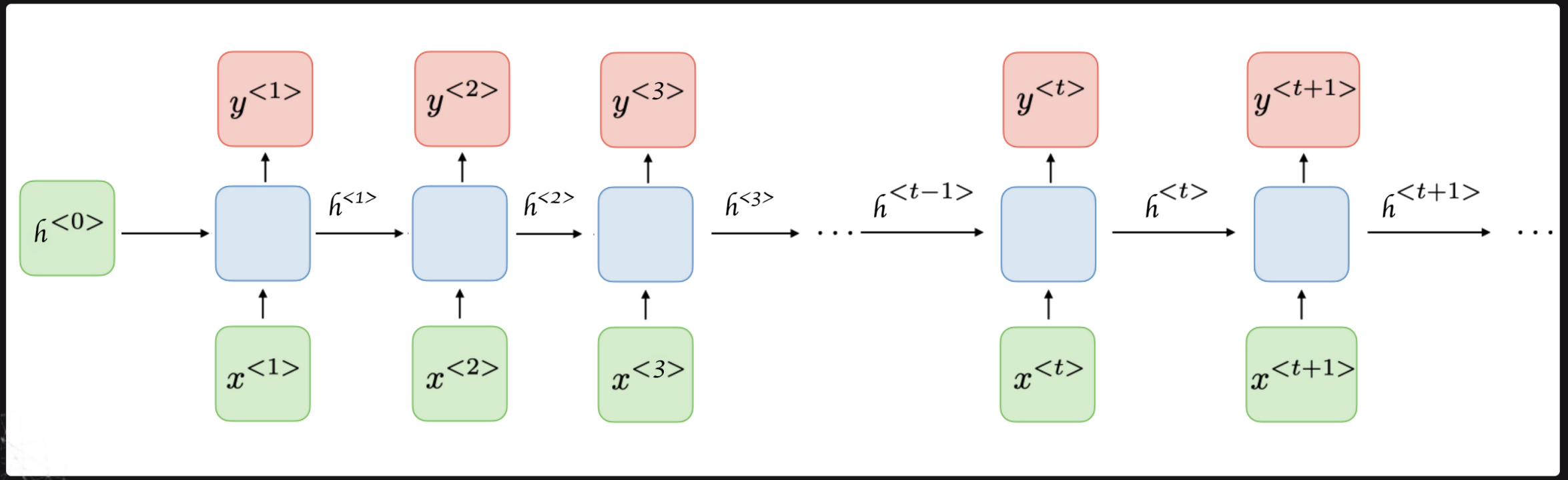
Recurrent Neural Network



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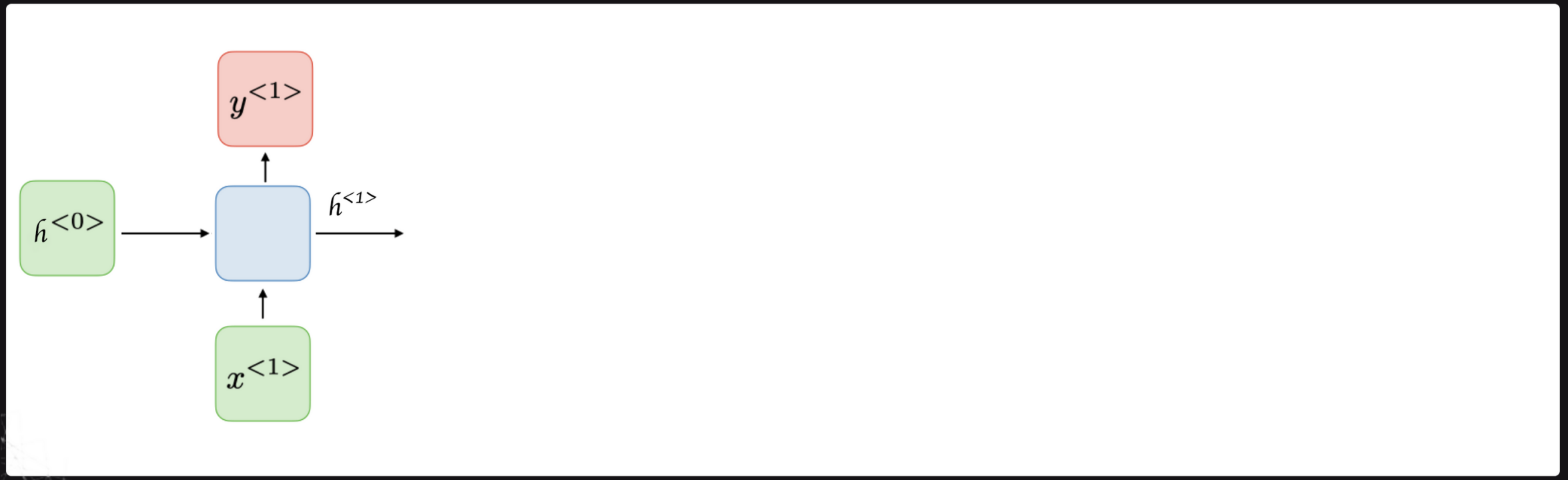
Recurrent Neural Network



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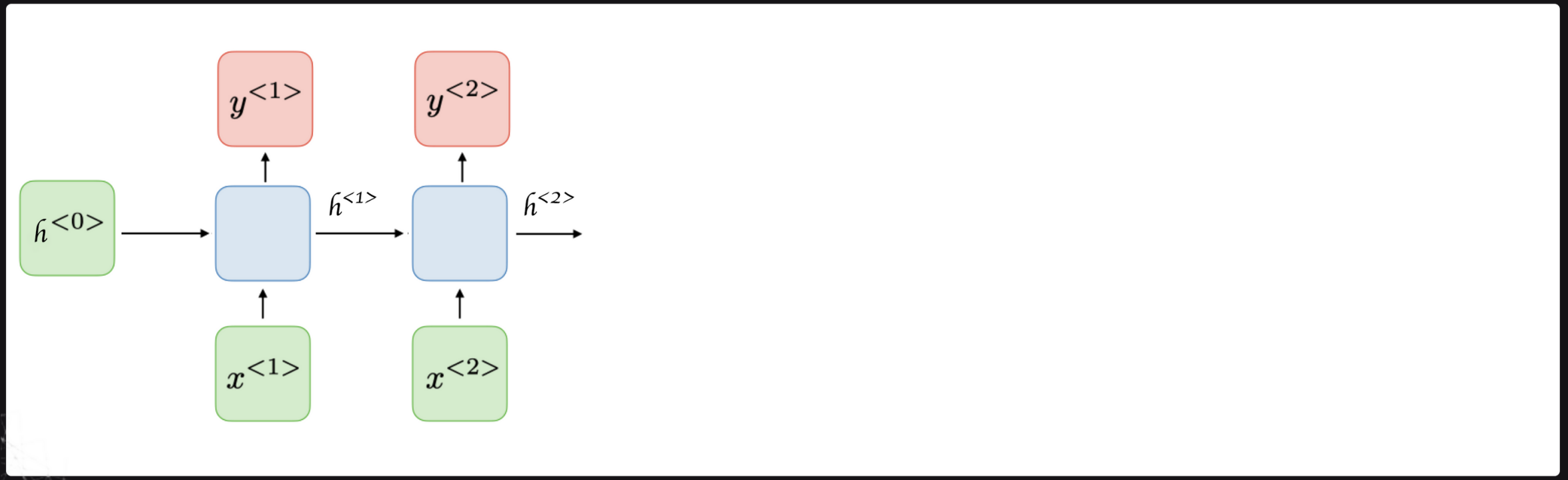
Recurrent Neural Network



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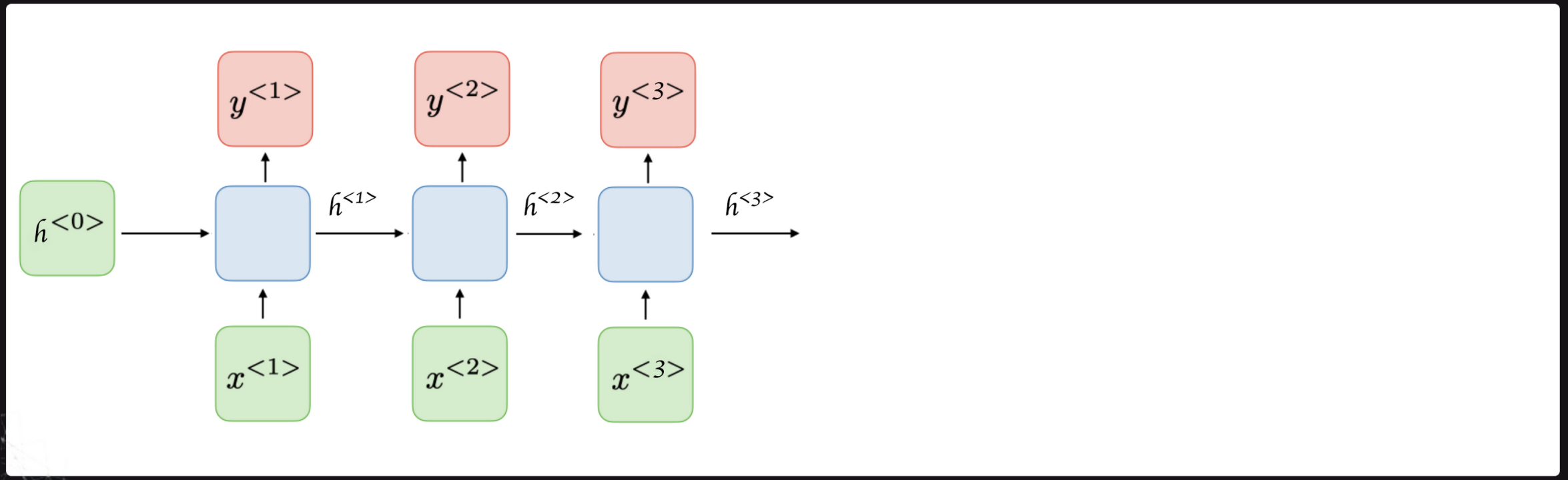
Recurrent Neural Network



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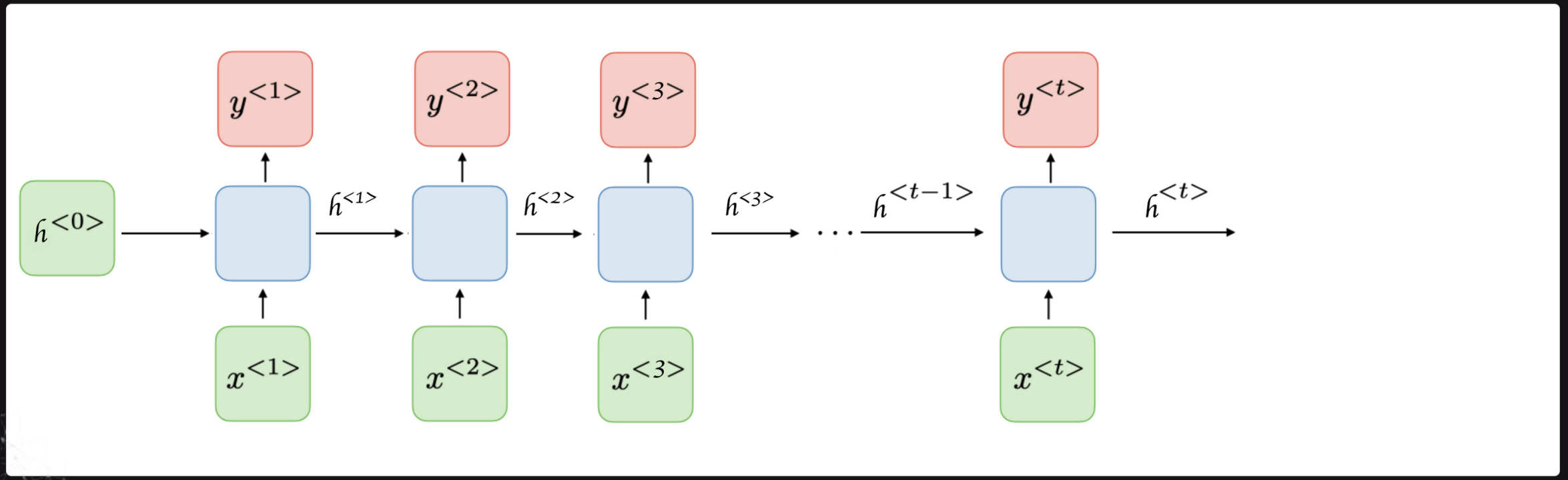
Recurrent Neural Network



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Challenge: Lower performance while working with longer sentences

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Vanishing/Exploding Gradients

Sepp Hochreiter observed that RNNs suffer from the vanishing/exploding gradients

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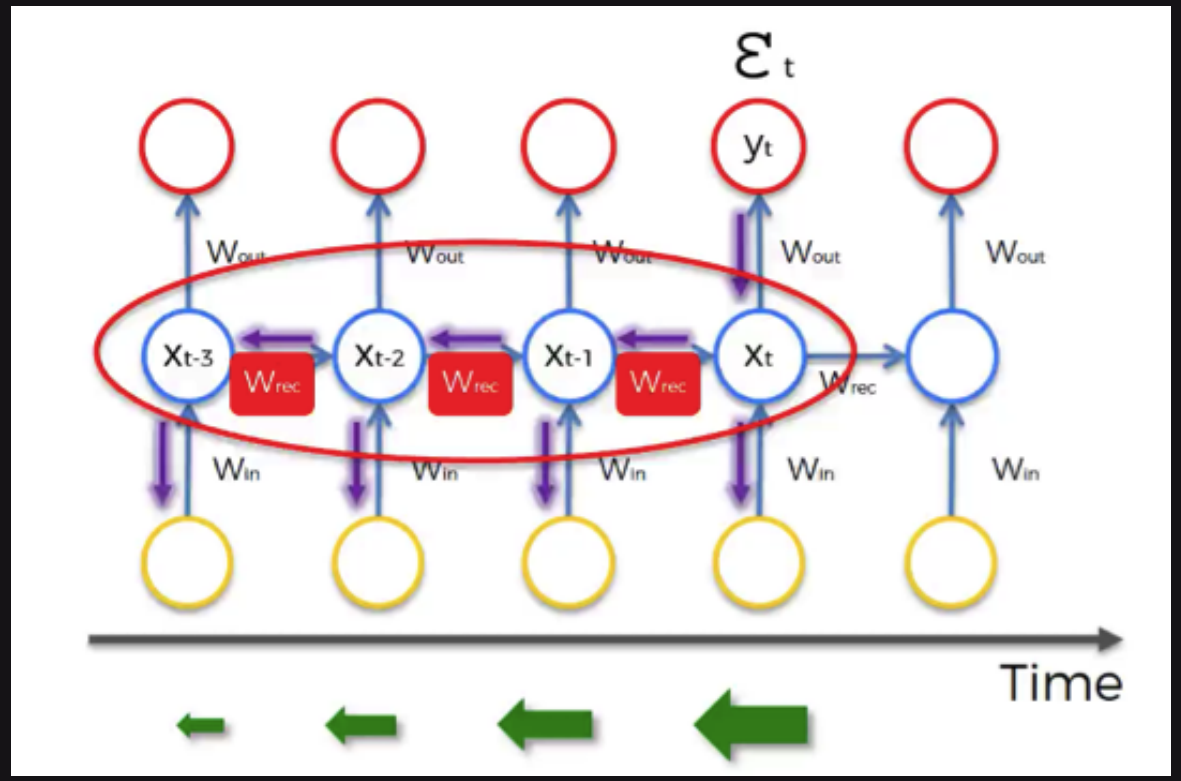
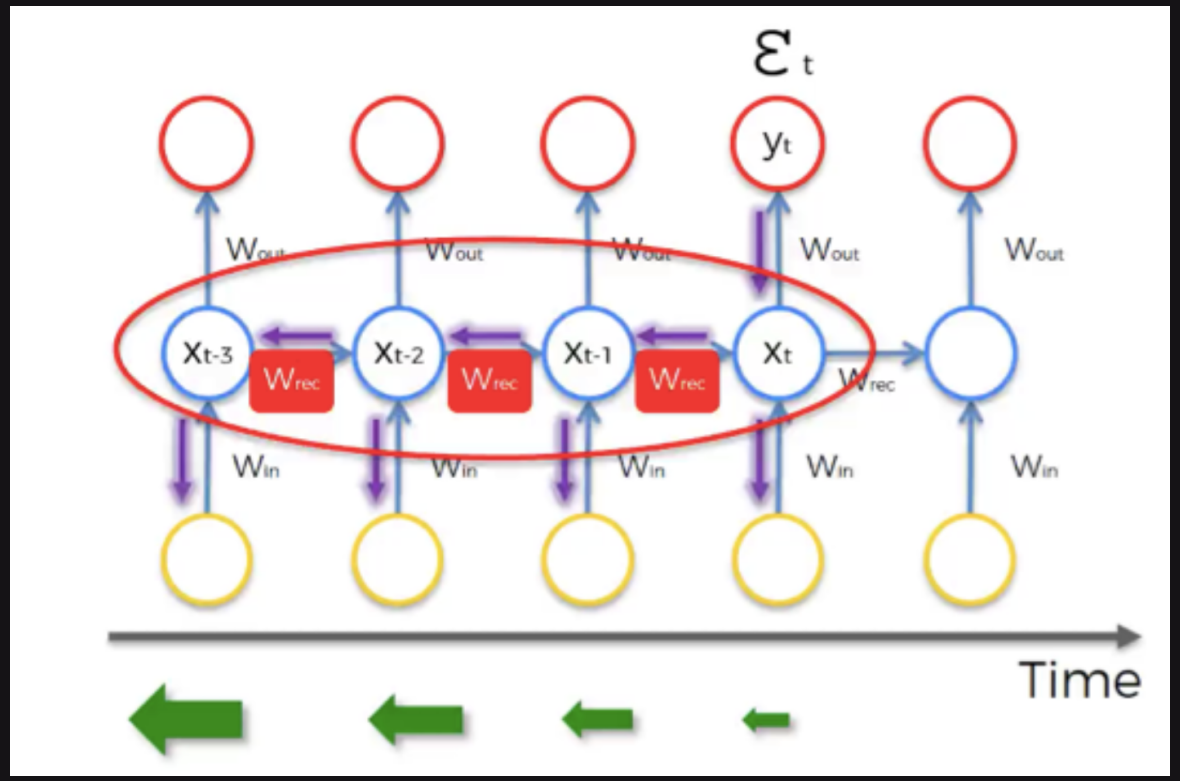
Vanishing/Exploding Gradients



Vanishing Gradient



Exploding Gradient

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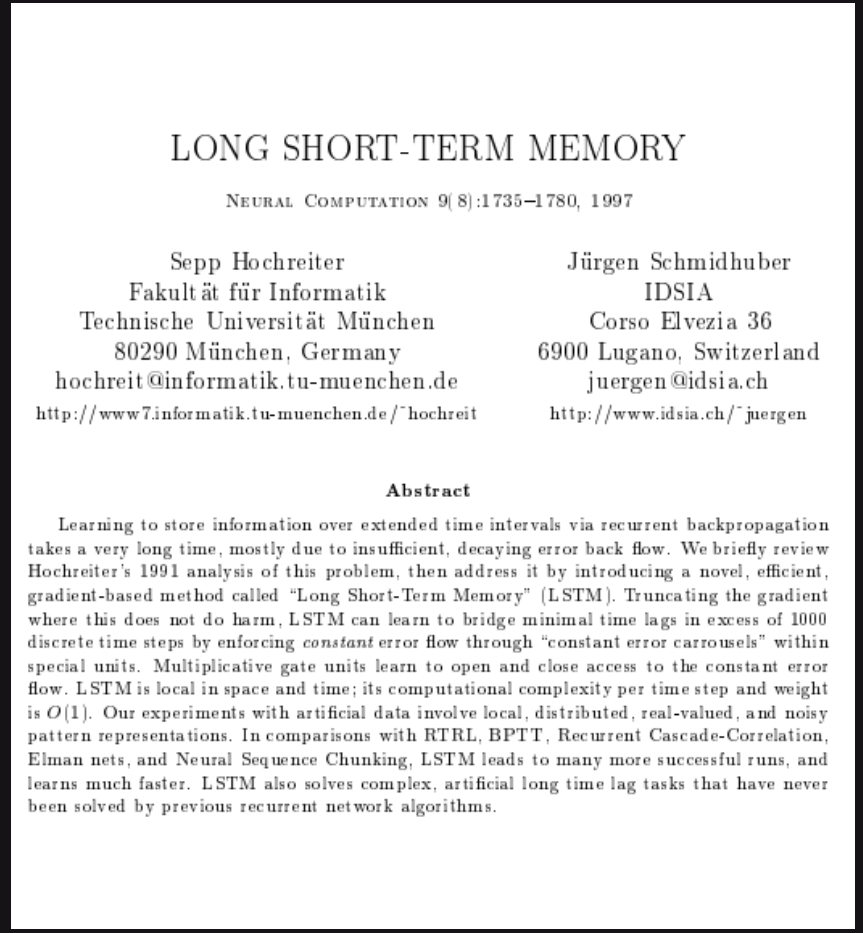
LSTM

- Most cited paper of 20th century!
- Overcomes Vanishing/Exploding Gradient through Gating mechanism

  Enables gradients to become stable and flow easily



LONG SHORT-TERM MEMORY

Neural Computation 9(8):1735–1780, 1997

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Abstract

Learning to store information over extended time intervals via recurrent backpropagation takes a very long time, mostly due to insufficient, decaying error back flow. We briefly review Hochreiter's 1991 analysis of this problem, then address it by introducing a novel, efficient, gradient-based method called "Long Short-Term Memory" (LSTM). Truncating the gradient where this does not do harm, LSTM can learn to bridge minimal time lags in excess of 1000 discrete time steps by enforcing *constant* error flow through "constant error carrousels" within special units. Multiplicative gate units learn to open and close access to the constant error flow. LSTM is local in space and time; its computational complexity per time step and weight is $O(1)$. Our experiments with artificial data involve local, distributed, real-valued, and noisy pattern representations. In comparisons with RTRL, BPTT, Recurrent Cascade-Correlation, Elman nets, and Neural Sequence Chunking, LSTM leads to many more successful runs, and learns much faster. LSTM also solves complex, artificial long time lag tasks that have never been solved by previous recurrent network algorithms.

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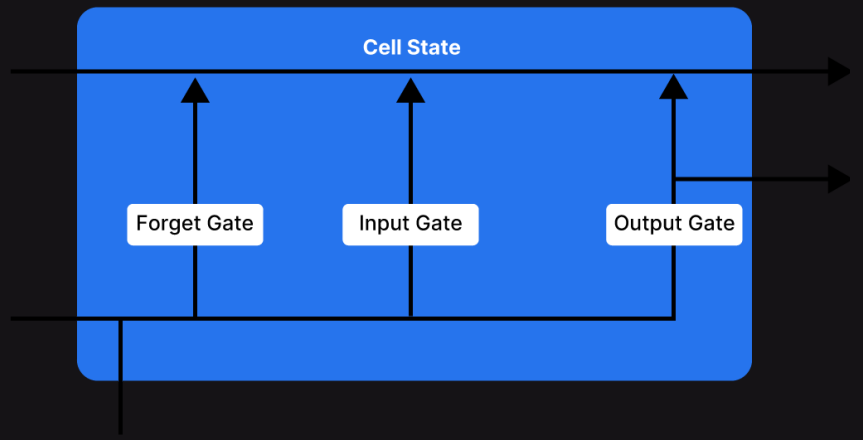
LSTM

- Most cited paper of 20th century!
- Overcomes Vanishing/Exploding Gradient through Gating mechanism

  Enables gradients to become stable and flow easily



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Neural Language Model

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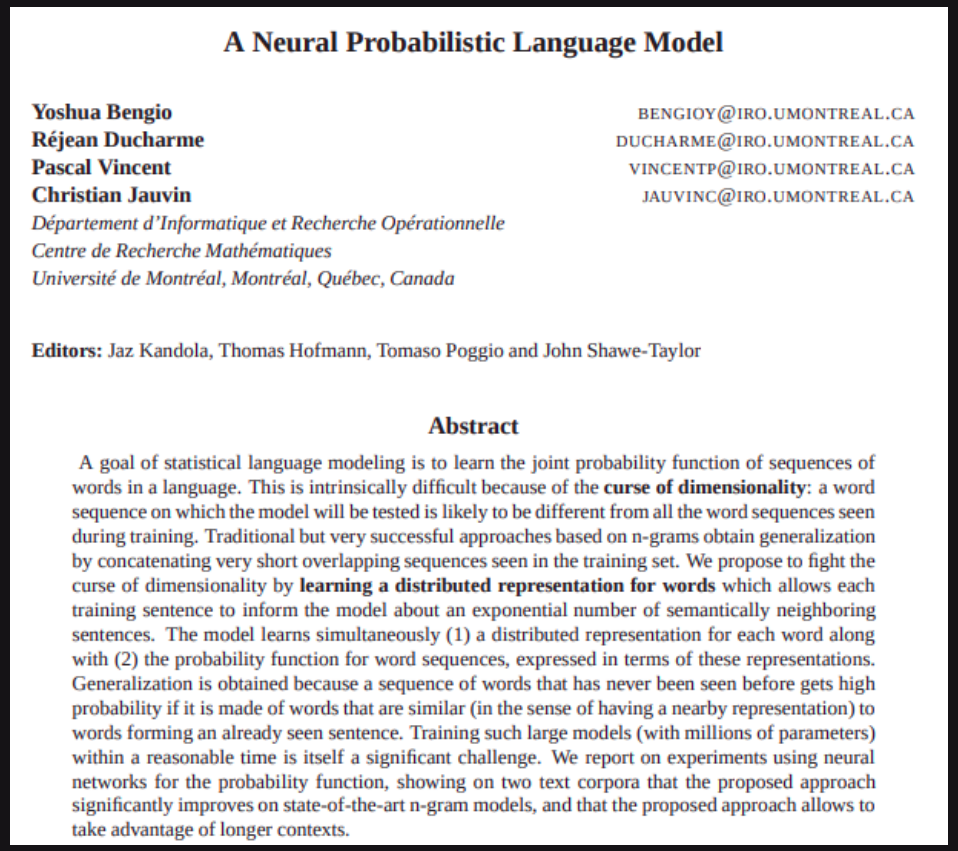
Neural Language Model



A Neural Probabilistic Language Model

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Abstract

A goal of statistical language modeling is to learn the joint probability function of sequences of words in a language. This is intrinsically difficult because of the **curse of dimensionality**: a word sequence on which the model will be tested is likely to be different from all the word sequences seen during training. Traditional but very successful approaches based on n-grams obtain generalization by concatenating very short overlapping sequences seen in the training set. We propose to fight the curse of dimensionality by **learning a distributed representation for words** which allows each training sentence to inform the model about an exponential number of semantically neighboring sentences. The model learns simultaneously (1) a distributed representation for each word along with (2) the probability function for word sequences, expressed in terms of these representations. Generalization is obtained because a sequence of words that has never been seen before gets high probability if it is made of words that are similar (in the sense of having a nearby representation) to words forming an already seen sentence. Training such large models (with millions of parameters) within a reasonable time is itself a significant challenge. We report on experiments using neural networks for the probability function, showing on two text corpora that the proposed approach significantly improves on state-of-the-art n-gram models, and that the proposed approach allows to take advantage of longer contexts.

Neural Language Model overcomes the limitations of the N-gram language models

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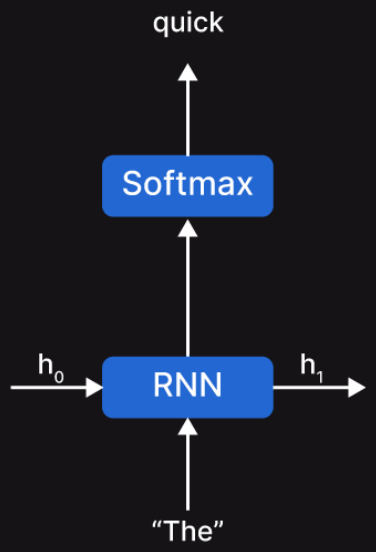
Neural Language Model



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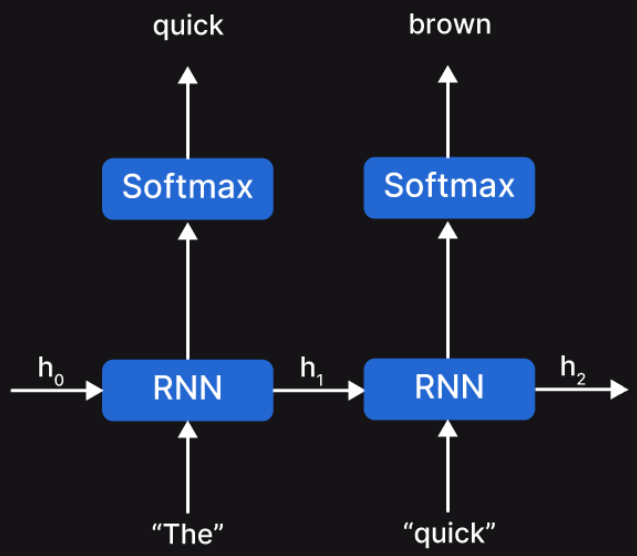
Neural Language Model



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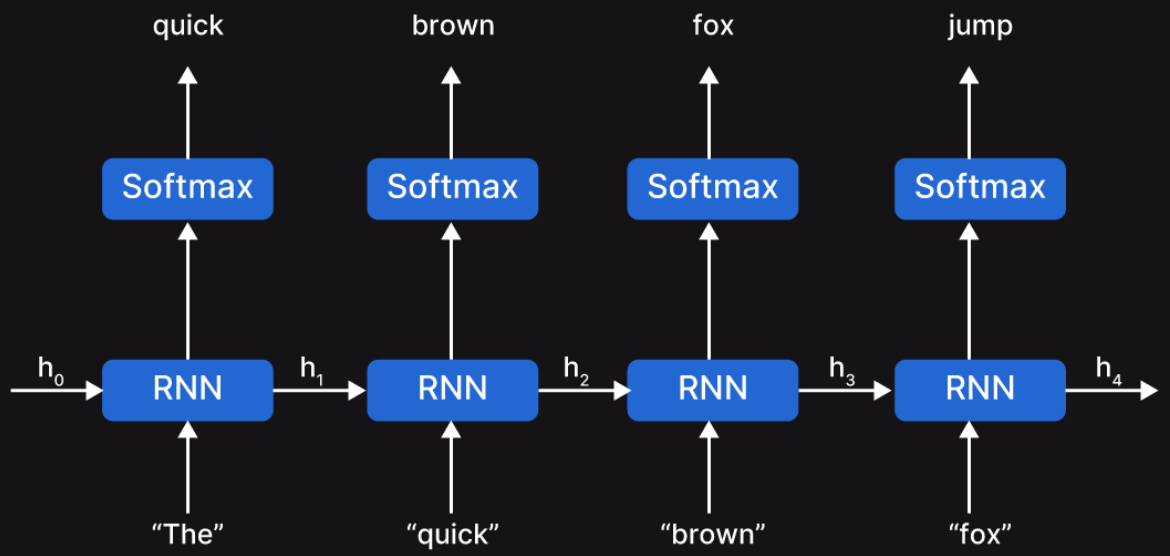
Neural Language Model



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The Evolution of NLP: Deep Learning Era II

Instructor

Kunal Jain

Founder and CEO, Analytics Vidhya

2014

Encoder Decoder

Architechture

Encoder Decoder architecture introduced for Seq2Seq modelling for the very first time in 2014

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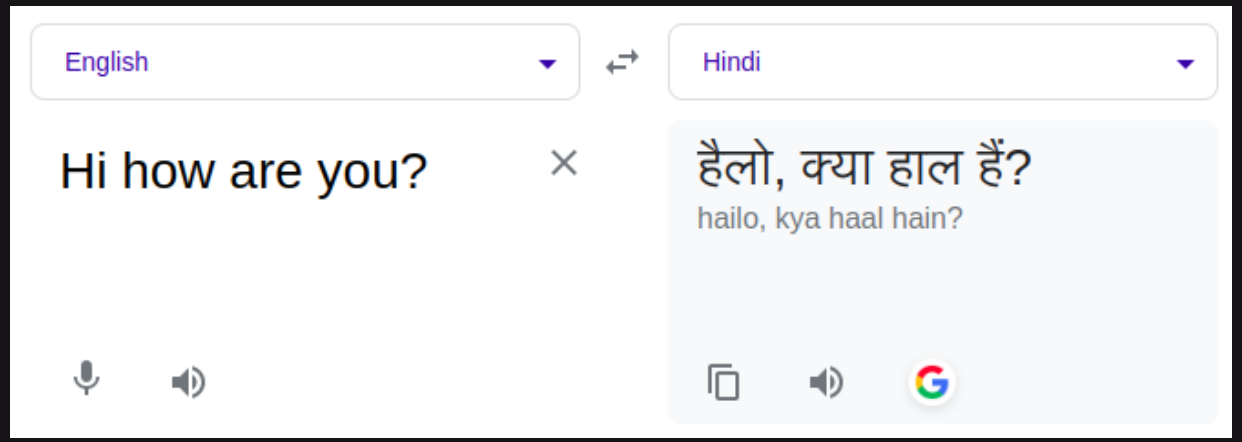
Encoder Decoder Architecture (2014)

- Seq2Seq model
- Inputs and outputs involving sequences of varying lengths



Machine Translation

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High level overview of Seq2Seq model

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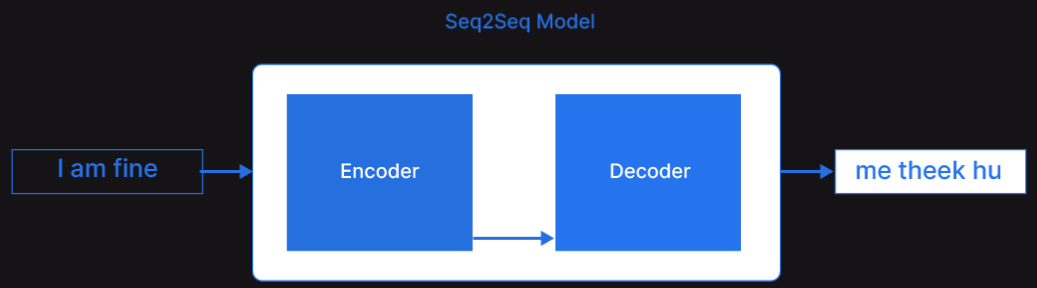
Seq2Seq model: Encoder and Decoder Architecture



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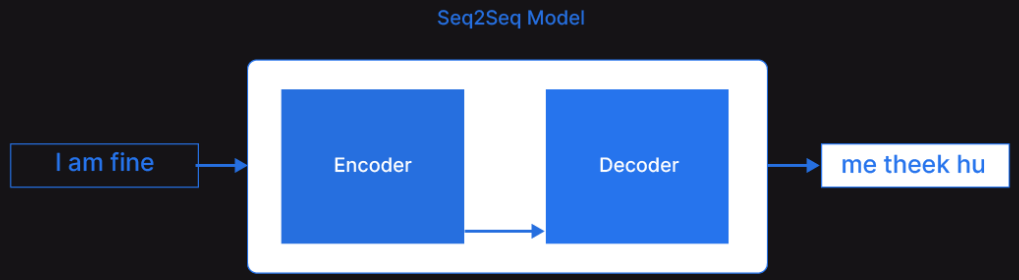
Seq2Seq model for Machine Translation



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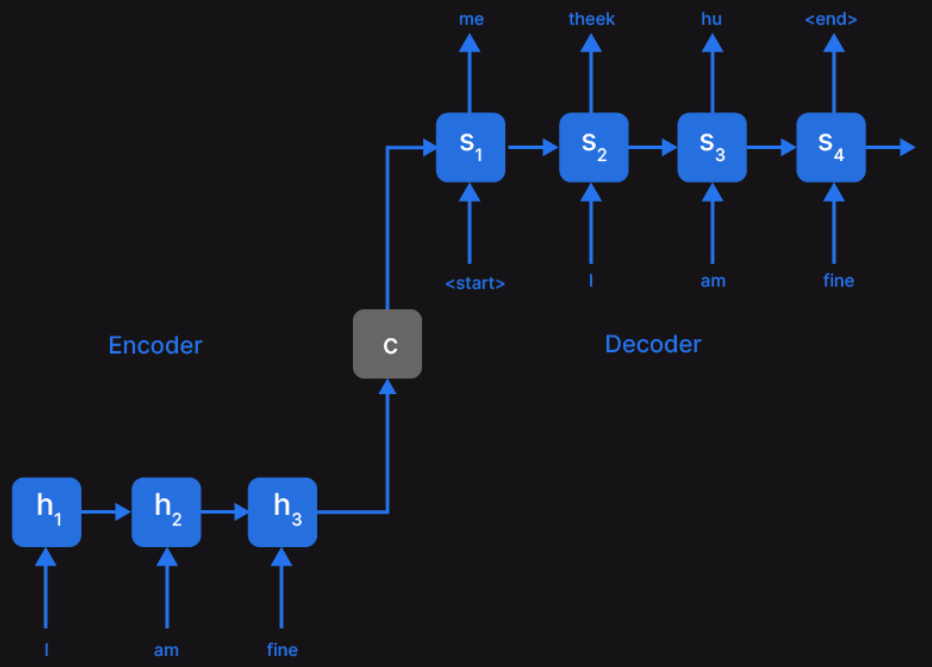
Encoder Decoder Architecture



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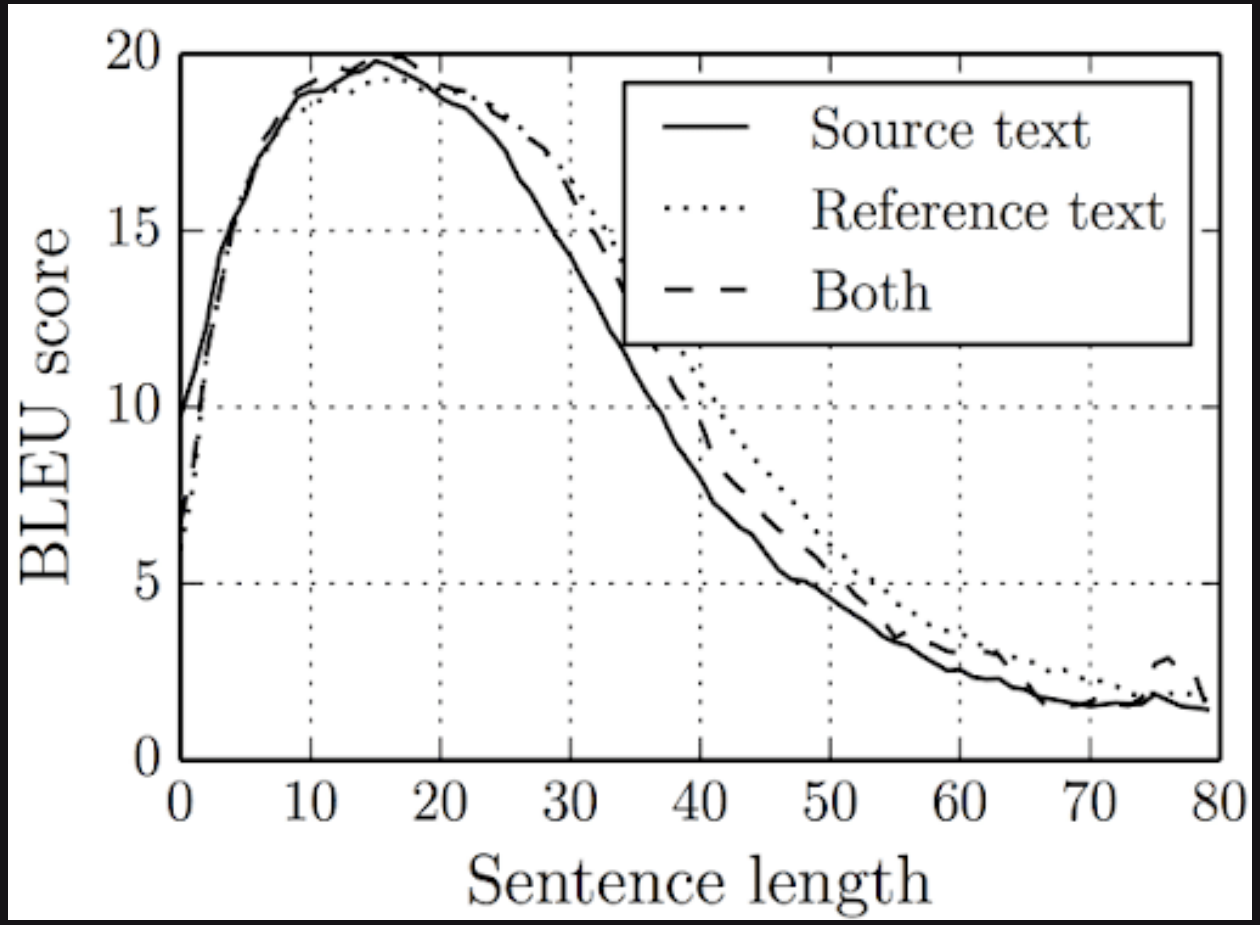
The performance decreases as the length of sentences increases



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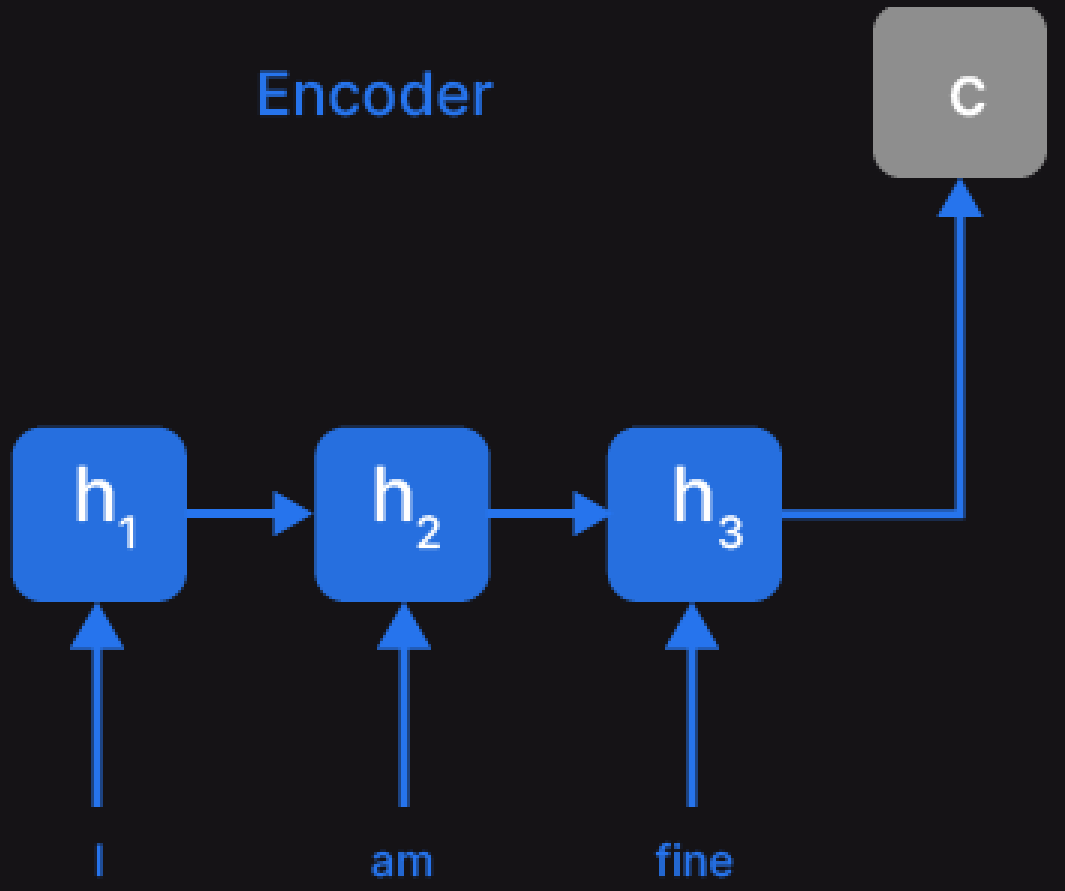
The reason being the encoder component



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2014

Encoder Decoder Architechture

Attention Mechanism solves the problem of Encoder Decoder architecture

2015

Attention Mechanism

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Attention mechanism focuses on certain parts of the input sentence while generating each word in the decoder.

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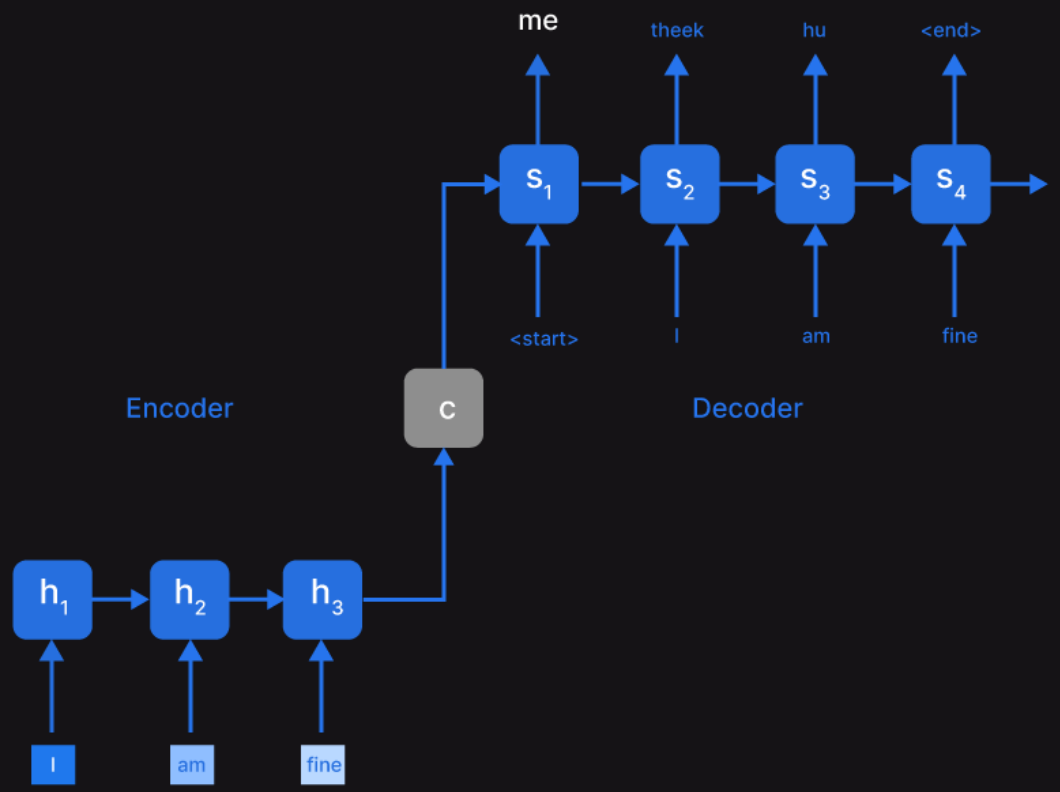

Attention Mechanism

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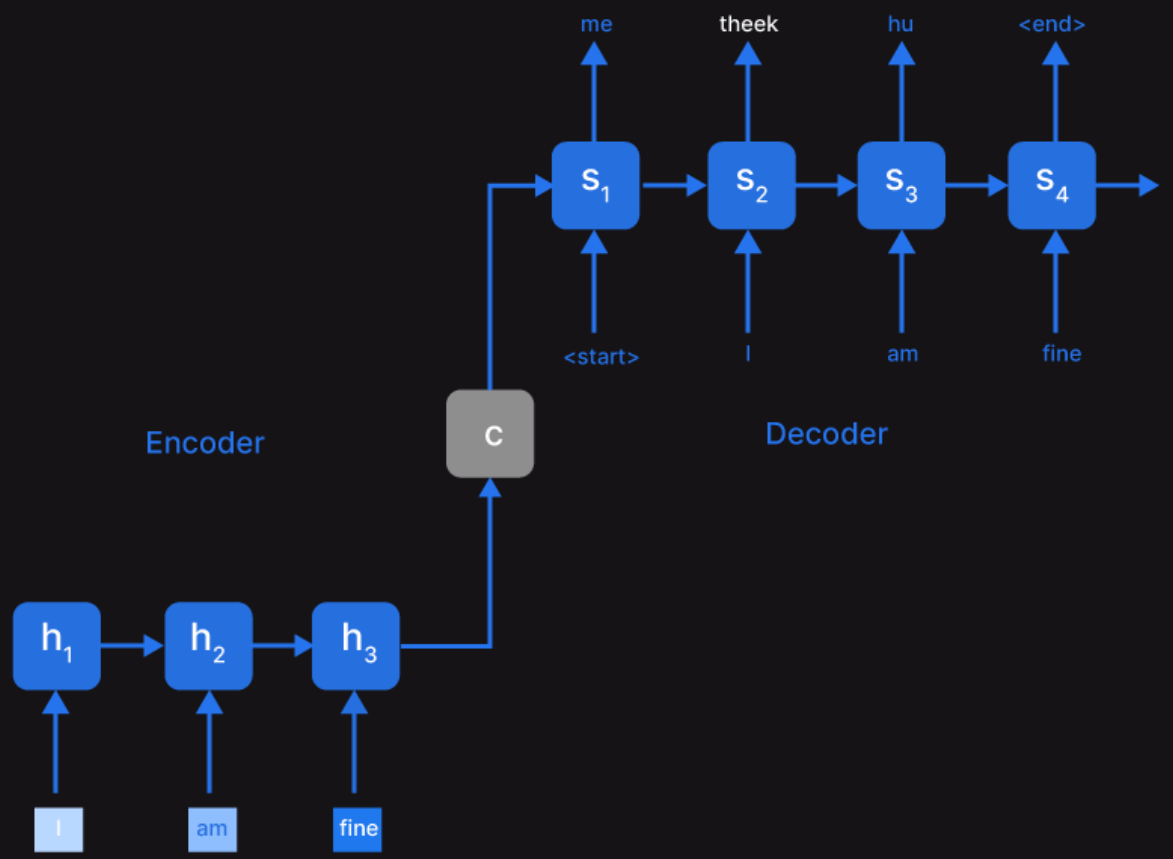

Attention Mechanism

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Limitation: Requires longer training time as there is no chance for parallelization.

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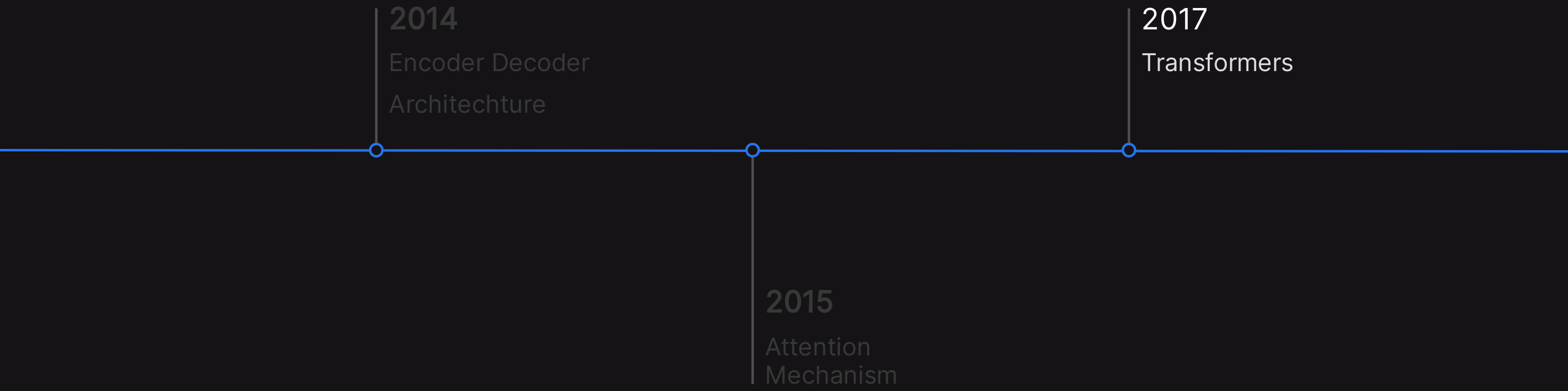
The Evolution of NLP: Transformers and Evolution

Instructor

Kunal Jain

Founder and CEO, Analytics Vidhya



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Transformers - The Beast of NLP

Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

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Augustus Odena

David Luan

Erich Elsen

Fred Bertsch

Kelsey Schroeder

Maxwell Nye

Niki Parmar

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LSTMs/GRUs are no longer needed!
Transformers = FFN + Attention mechanisms

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High Level Design of Transformers

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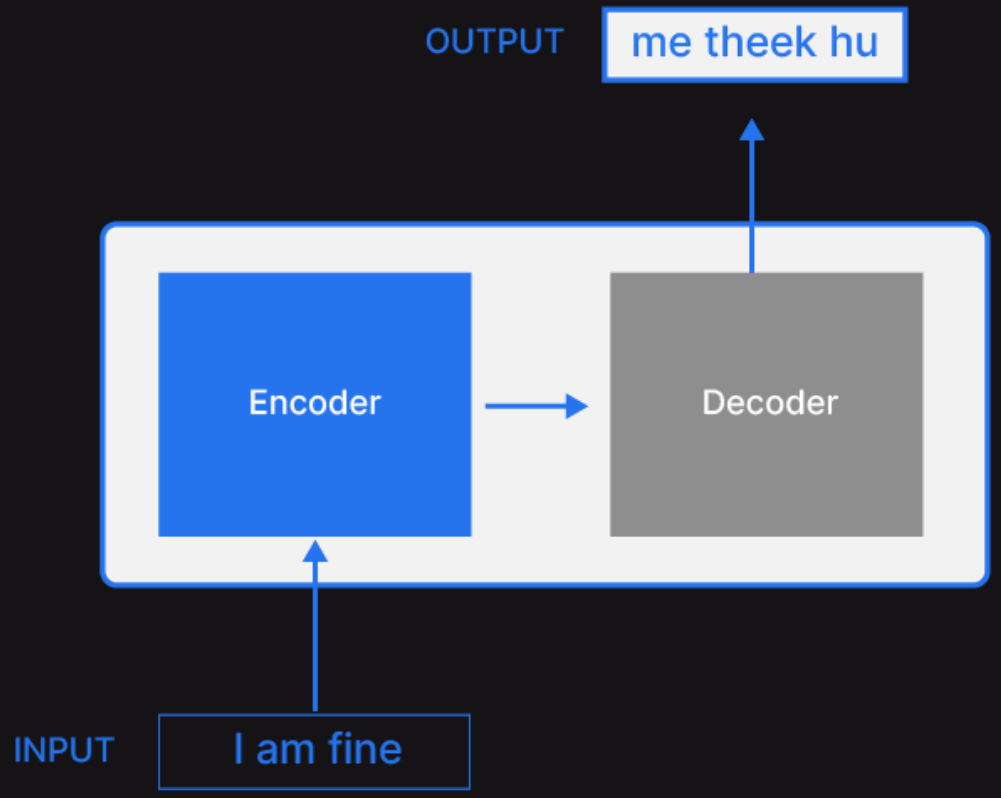

High Level Design of Transformers

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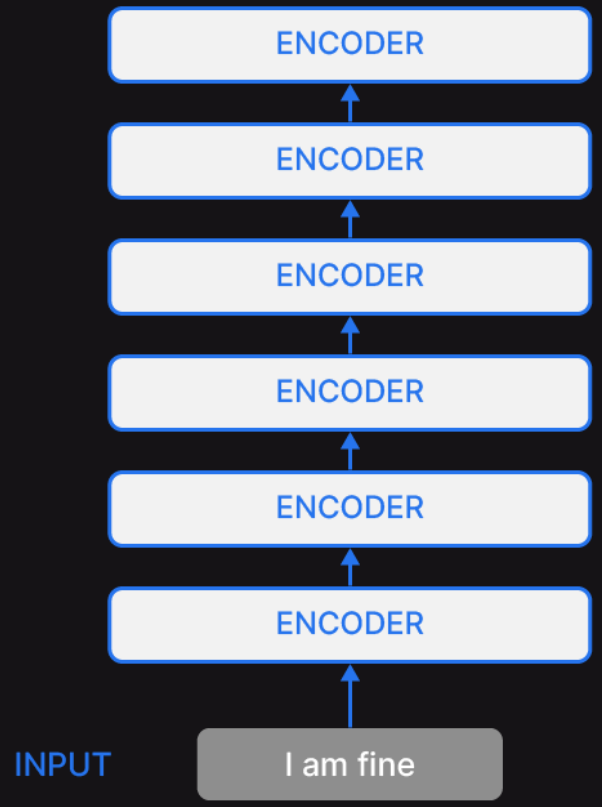

Encoder Component

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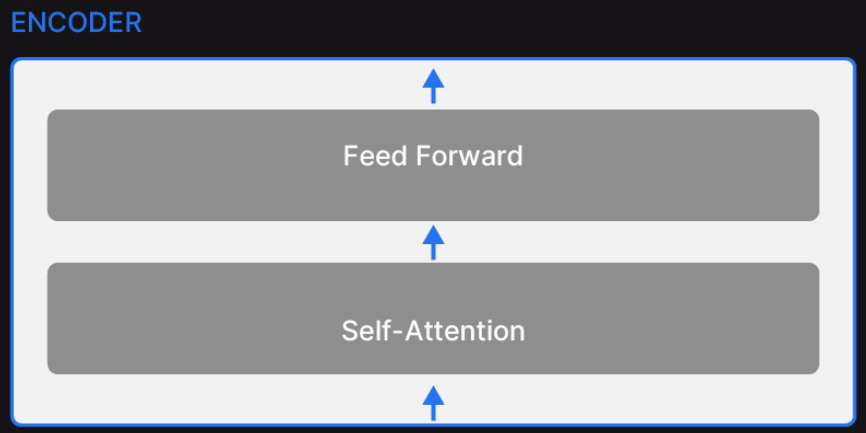

Encoder Block

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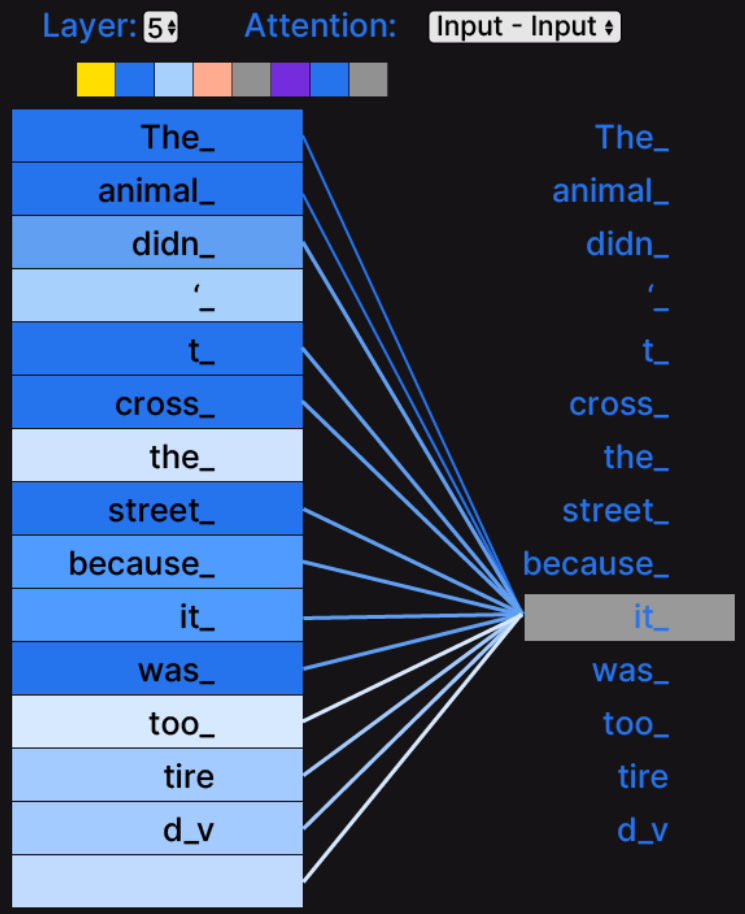
Self Attention



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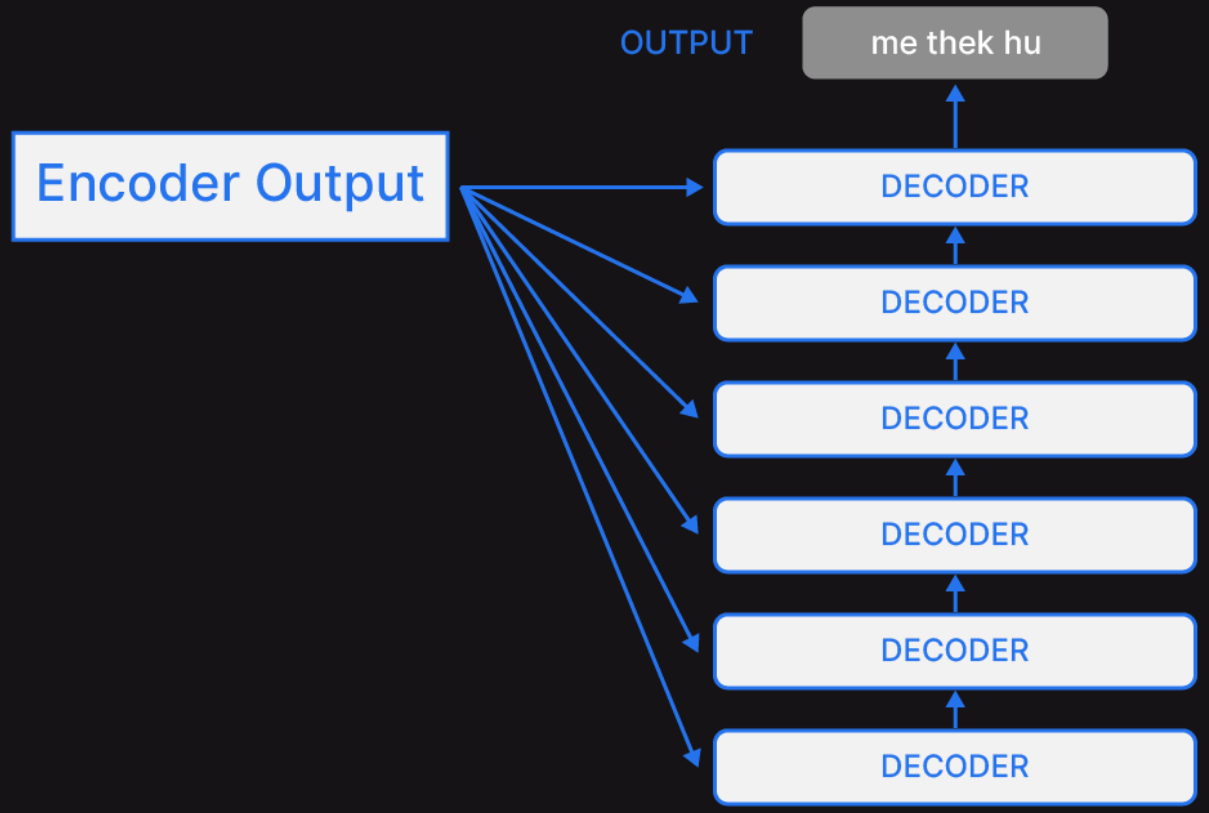

Decoder Component

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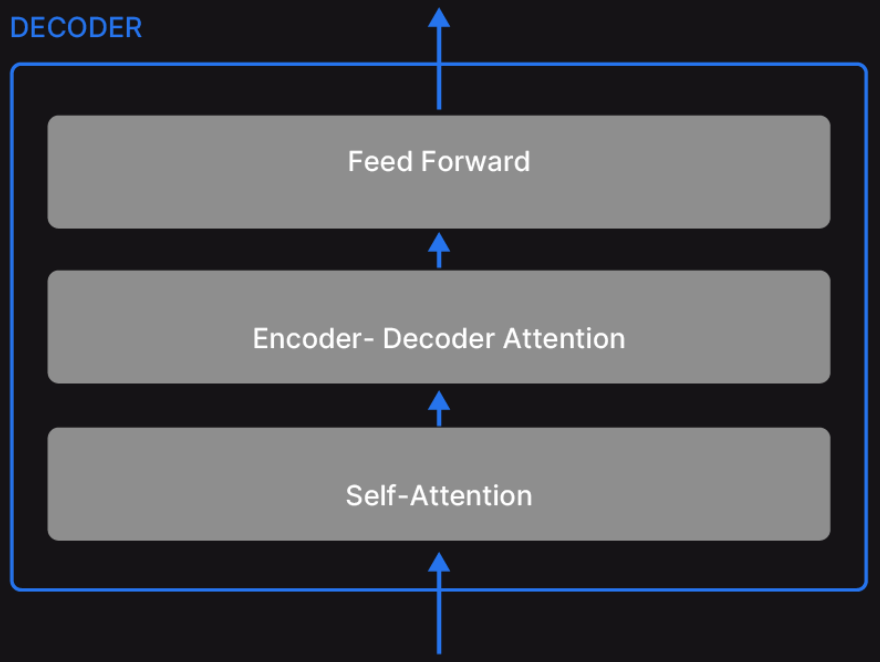

Decoder Block

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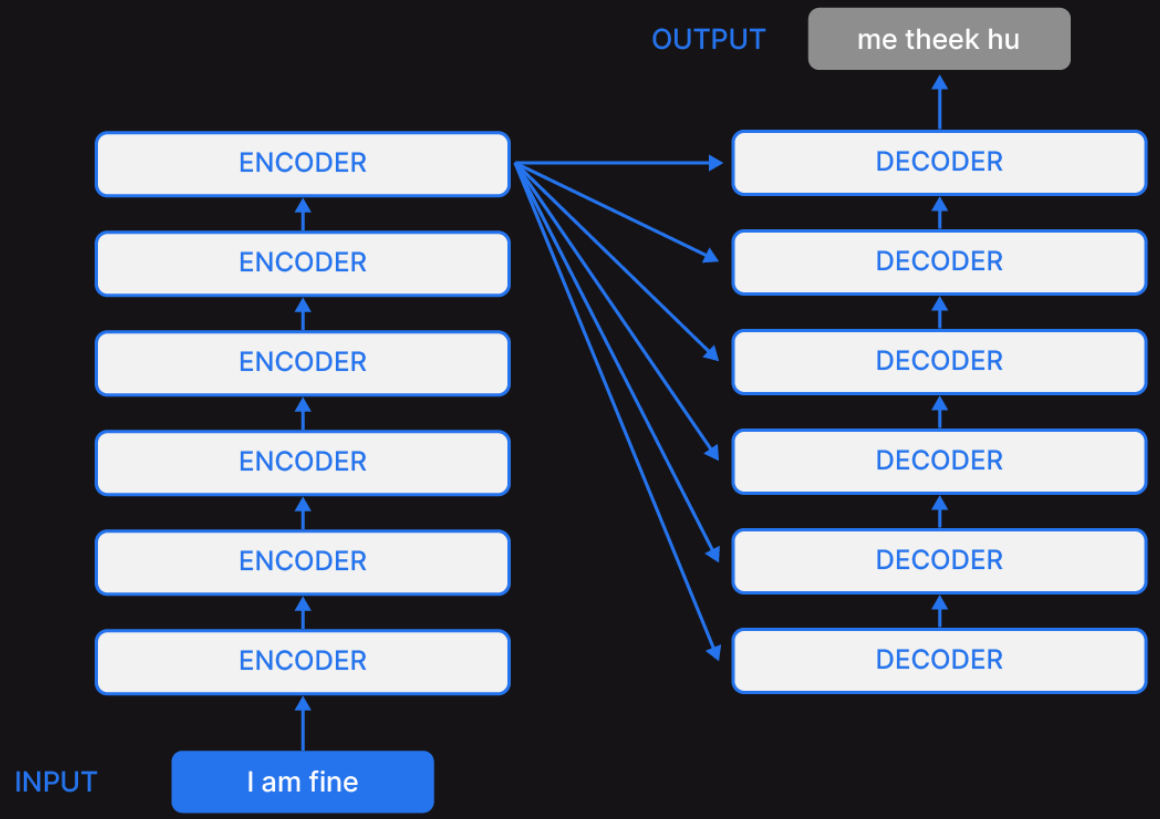

Transformers

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Transformers → The Era of Large Language Models

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